

Evaluation of energy-saving potential for optimal time response of HVAC control system in smart buildings

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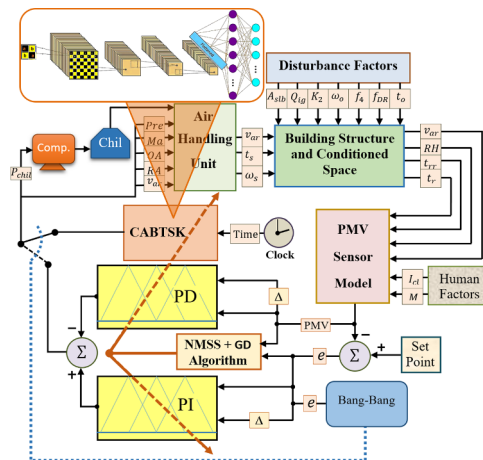
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HIGHLIGHTS

- A novel integrated two different types to control a large-scale nonlinear system.
- Converting Takagi-Sugeno fuzzy inference system into faster hybrid layers structure.
- Spans of the fuzzy membership functions are tuned on-line to achieved energy saving.
- Online tuning by downhill simplex algorithm achieved minimum time response.
- Integrated control deal with plants possess lag time, big inertia and uncertainty.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Boolean logic
Fuzzy PI-PD controller
Fuzzy TSK controller
Integrated controllers
MIMO HVAC systems

ABSTRACT

In some fields, such as the semiconductor manufacturing process, museum, pharmaceutical, and medicine manufacturing industry, the HVAC system needs a very fast response time to protect products and more energy-efficient buildings than traditional controllers. So, the proposed controller is designed to overcome such problems by using integrated fuzzy PI-PD Mamdani-type (FPIPD) and cluster adaptive training based on Takagi-Sugeno-Kang (CABTSK) type. The spans of the fuzzy membership functions of the FPIPD are tuned online by the Nelder-Mead simplex search (NMSS) algorithm to minimize time response, while the CABTSK model is tuned offline and online using a gradient descent (GD) algorithm to enhance the stability of the overall system and reject disturbances. Then, the integration framework is used to enable the concept of time-optimal based on the bang-bang code delegation. In this sense, a selected switch delegates the execution of proper control code to the action processor that provides computational resources to control indoor conditions. The predicted mean vote (PMV) index provides a higher comfort level than the temperature, as it considers six variables related to thermal comfort. The results of the proposed structure show that it improves the overall output accuracy and significantly reduces the response time. Furthermore, it increases the robustness of the indoor conditions and it is

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quite applicable to the MIMO HVAC systems processes with strong coupling actions between temperature and humidity, large time delay, noise, disturbances, nonlinearities, and imprecise identification model.

Nomenclature

Symbols

A	surface area, m ²
C	heat capacitance, J/°C
$\frac{dE_s}{dt}$	rate of change in storage energy of the system, J/s
$\dot{E}_{in}$	energy rate entering the system, J/s
$\dot{E}_{out}$	energy rate leaving the system, J/s
M	mass, kg
C _p	specific heat, J/kg·°C
$\dot{m}$	mass flow rate, kg/s
M _{cp}	heat capacitance, J/°C
T	temperature, °C
ω	humidity ratio, kg _w /kg _{da}
h	latent heat/ heat transfer coefficient, J/kg, W/(m ² ·°C)
$\dot{Q}$	cooling load, W
CF	surface cooling factor, W/m ²
U	construction U-factor, W/(m ² ·°C)
ΔT	cooling design temperature difference, °C
OF _t , OF _b , OF _r	opaque-surface cooling factors
DR	cooling daily range, °C
	surface cooling factor, W/m ²
U _{NFRC}	fenestration U-factor, W/(m ² ·°C)
PXI	peak exterior irradiance, W/m ²
SHGC	solar heat gain coefficient
IAC	interior shading attenuation coefficient
FF _s	fenestration solar load factor
E _t , E _b , E _D	peak total, diffuse, and direct irradiance, W/m ²
T _x	transmission of the exterior attachment
F _{shd}	fraction of the fenestration shaded by overhangs or fins
L	site latitude, °N
SLF	shade line factor
D _{oh}	depth of the overhang, m
X _{oh}	vertical distance from the top of the fenestration to the overhang, m
F _{cl}	shade fraction closed (0 to 1)
K	turbulent kinetic energy (m ² /s ²)
RNG	Renormalization group
ψ	exposure (surface azimuth), measured as degrees from the south
$\dot{V}$	volumetric flow rate, L/s

DF	infiltration driving force, L/(s·cm ²)
$\mathfrak{R}$	thermal resistance, °C/W
N _{oc}	number of occupants
N _{br}	number of bedrooms
α_{roof}	roof solar absorptance
τ	time constant, s
I	infiltration coefficient
$\Delta\omega$	indoor-outdoor humidity ratio difference, kg _w /kg _{da}

Subscripts

m	air in a mixing box
r	room/ return
o	outside
os	outside supply
i	inside
He	heat exchanger
a	the air
w	water
aHe	air in the heat exchanger
L	leakage
W _{in}	water input
W _{out}	water output
Wl	wall
room	inside room
out	outside room
g	glass
fg	heat of vaporization
Opq	opaque
inf	infiltration
fen	fenestration
f	indoor and outdoor
t	at time t
flue	flue effective
es	exposed
ul	unit leakage
ig	internal gains
l	latent
s	sensible/ supply
fur	furniture
cl	closed

1. Introduction

The major challenge facing the world today is the greenhouse gas (GHG) emissions [1]. It is reflected in the fact that over one-third of the total CO₂ emissions caused by electricity production [2]. The building sector uses about 74% of the electricity consumption, and more than 56% of this consumption was imputed to the HVAC (heating, ventilation, and air-conditioning) systems [3]. These require considerable efforts to improve the overall energy efficiency of buildings. The HVAC control systems are one of the most relevant parts in terms of energy-saving and environmental protection in the building sector [4,5]. The development of HVAC control strategies led to saving of 60% in its power consumption [6,7]. The integrated HVAC system control based on a dynamic optimization technique has profound significance on energy saving and energy costs due to reductions in peak load as well as improvement of indoor conditions [8,9]. Therefore, time of a control

system response is complemented energy saving measures especially for operation in indoor conditioned spaces [10].

HVAC control system concerns stability, accuracy, sensitivity, reliability, and response time aspects; among these issues, the last theme is the less argued in literature and this paper focuses on the fast response time and reduces energy consumption by HVAC systems. Therefore, based on these aspects, the primitive bang-bang (on-off) controller is characterized where it never settles in one state (the system continually overshoots the reference state or hunting) as its manipulated variable is adjusted from fully closed to fully open. Accordingly, it gradually become obsolete owing to rapid development in control algorithms and to the fact that it includes a hunting or a cycle around the set point value, delay time and hysteresis [11]. Therefore, adjusted proportional gain in the process loop of this type of controller is used close to the set point to prevent hunting. This leads to continuously adjust its parameters manually to increase the control sensitivity,

response and reduce hunting, and this is impractical for control applications, beside its oscillation cycle cause extensive wear to the control actuators. Sometimes, compressor on/off cycling is used in the HVAC systems to take advantage of the compressor which is a prohibitive cost when controlled by continuously variable speed [12,13].

Most of the recent studies ignore the concerns of fast response time regarding the influence of conditioned place service, where some services are involving the processing and storage of products. For instance, the study [14] establishes that the intelligent controller can be accurate and achieved energy saving but it did not concern with issues related to optimal time response. Another study [15] dealt with energy efficiency and thermal comfort in buildings and it did not have too much consideration for the desired response-time characteristics. A further study [16] has investigated the energy recovery ventilator based on the PID controller to assist the Gray-box models of the residential HVAC system, but that is insufficient due to a lack of control ability to respond fast and effectively. The PID control algorithms are the most popular approach for HVAC system process control, despite the development of PID algorithm research in continuous improvement, energy integration, subsystem unification and requirements for high product quality have resulted in pure lag times and non-linearity for renewed buildings and HVAC systems [17]. However, several possible techniques are available to reduce effects of some of the dominant shortcomings for the PID status to tackle the complex issues that hound the MIMO processes types such as modern HVAC systems [18].

For such system, the use of PID to manipulate inputs system with a steady set parameter will provide poor performance, which means that it will not control indoor conditions to provide thermal comfort [19]. For this reason, the PID needs to monotonously tune the adjustable parameter to get superior performance for the PID, which becomes inapplicable to such systems. Therefore, the online tuning of the PID parameters was theoretically obtained by using fuzzy sets for the same HVAC system. Optimization through the use of the fuzzy modeling approach of PID parameters was performed to maximize the performance of the system [20]. Soyguder and Alli [21] employed PID in the ANFIS controlled method for HVAC system to handle plant nonlinearity to control indoor temperature and relative humidity; however, they were not concerned for solving the coupling problem of the relative humidity and the indoor temperature. In previous work, it solved the coupling problem with less energy consumption by using predicted mean vote (PMV) [10]. Where, Jiang et al. [22] solved the decoupling system by employing PI in ANFIS control in order to manipulate the two input variables of the plant (air volume and water flow that are responsible for controlling temperature and relative humidity); even though, they did not consider whether the outdoor condition was one of the high relative humidity. Regarding this case, the HVAC control system can only cope with this problem by adding pre-cooling coils or reheating coils [23], and the reheating coils are not recommended when trying to save energy. Furthermore, previous studies [24] did not compare the backpropagation algorithm of online tuning performance to other algorithms, even though the backpropagation algorithm has certain deficits (e.g., the backpropagation algorithm suffers from two main drawbacks: network paralysis (forcing the neurons to operate at large outs) and trapping at the local minima [25]).

The minimum time required to get set point of indoor conditioned space by fuzzy controller is achieved by tuning the fuzzy logic for HVAC control system. Such controller is known as self-tuning fuzzy logic controller (STFLC) besides it can be further classified as an indirect or direct self-tuning. Accordingly, the STFLC mechanism provides dynamic gain by adjustment of the parameters to get superior performance for the HVAC control system. Recently, online tuning algorithm has successfully achieved many nonlinear system control objectives by using different algorithm approaches [26]. Most intelligent controls are tuned by the gradient descent (GD) and Nelder-Mead simplex search algorithm, also known as downhill simplex algorithm or amoeba algorithm, abbreviated as NMSS. The first one, GD is the most widely

used as it is faster than the fastest one known as Levenberg-Marquardt (LMB) algorithm [5]. As for the second one, its application is done without the need for gradient, which is preferred, as it does not use numerical or analytical gradients and it can be applied to a wide range of objective functions and membership function; therefore, it is an efficient for direct search [27]. In addition to the NMSS is simple, and without derivatives, it can be programmed on a computer [28]. Furthermore, The NMSS algorithm is adopted for optimization, which leads to achieving a competitive advantage in the form of reducing response time.

The demand of comfort life will continue to grow as the complexity of structure of the control algorithms grows. In this paper, the two controllers fuzzy PI-PD Mamdani-type (FPIPDM) and cluster adaptive based on Takagi-Sugeno-Kang (CABTSK) are altered by bang-bang action. According to its objectives, besides considering the minimum time required to get a set point of indoor conditioned space, it considers precisely controlling the highest comfort level for building occupants. A time-optimal based on the bang-bang code is proposed to delegate decisions for which controller to execute control actions to manage the variables of an HVAC system. Integrating two different types of controllers is feasible to implement proper control approach to overcome the high nonlinear nature of the whole HVAC system. The FPIPDM controller is optimized by the NMSS algorithm, which leads to minimum response time for initial conditions including reset condition by varying the reference point of the PMV. The NMSS algorithm is a multivariable unconstrained and can accommodate PMV and PMV rate as input variables for controllers. When error signal passes the threshold parameter of the comparison algorithm, the algorithm immediately shifts a task into proposed controller to execute another controller, thus giving rise to various levels of control. The FPIPDM has four inputs and six outputs, where the CABTSK has one input and six outputs. Under these circumstances, the bang-bang algorithm can determine which control signal is suitable. The sequential self-tuning is quite simple to regulate multivariable and continues until both controllers converge to desired objectives. The tuning principle of FPIPDM is achieved by manipulating spans of the middle fuzzy sets of membership functions (MFs) for the four inputs, where time response depends on the span of the MFs [29]. Using GD for self-tuning of CABTSK is attained by tuning clusters' weights for each output.

This paper contributes to integrating two types of controllers, which leads to achieving a competitive advantage in the form of reducing response time. As such, this work proposes and develops the construct of an innovative HVAC system controller depending on harmonizing two types of controllers, whereby the main contributions can be briefly summarized as follows:

- 1- A novel approach to integrate FPIPDM and CABTSK into a common framework for the HVAC control system, which improves significantly the response time especially when getting started and eliminates the steady-state error.
- 2- The FPIPDM is designed to address the needs of minimizing the response time, as it is acquired by the online spans tuning of the membership function of the fuzzy set.
- 3- The mechanism of creating the CABTSK structure is based on converting the Takagi-Sugeno fuzzy inference system into a hybrid layers structure to deal with highly nonlinear systems.
- 4- On this base, hybrid layers are fabricated from hybrid vectors by the clustering concept of the learning data set.
- 5- The online tuning algorithms for both controllers (FPIPDM and CABTSK) are using NMSS and GD for nonlinear regression.

2. Specification of modeling techniques

The framework of this section will summarize the specification of model's characteristics and its requirements must meet real condition and be approved by our works before. The first modeling part is related

to the building and HVAC systems and the second part is associated with the PMV.

2.1. Integrated building and HVAC systems

To obtain a real behavior of the indoor model, this work suggests a more inclusive approach that includes all parameter values in the sub-models of integrated building and HVAC systems. As such, this model is characterized by MIMO type, nonlinear, large-scale, large thermal inertia, uncertain disturbance factors, time variability, nonlinear constraints and existing properties of outputs coupled. This model type is used as an example to demonstrate the control ability of large-scale system, which is adopted from the previous work [23]. For more details and descriptions of how the model was divided into five subsystems, it is defined in the previous work, which can be represented by Eq. (1).

$$\begin{array}{c} \text{TheBuildingStateEquation} \\ \left[\begin{array}{c} T_o(t) \\ k_2 \\ T_r \\ T_{rr} \\ f_{DR} \\ A_{slab} \\ k_3 \\ f_4 \\ \omega_o(t) \\ \dot{Q}_{ig,l} \end{array} \right] \\ \left[\dot{X} \right] = f(X, t) + G(X, t) \end{array} \quad \begin{array}{c} \text{TheWholeHVACStateEquation} \\ \left[\begin{array}{c} \dot{m}_{os} \\ \dot{m}_r \\ \dot{m}_w \\ T_o \\ T_{rr} \\ \dot{m}_{mw} \\ \omega_o \\ T_r \\ k_2 \\ A_{slab} \\ F_{Dr} \\ f_4 \\ \dot{Q}_{ial} \end{array} \right] \\ \left[\dot{X} \right] = f(X, t) + G(X, t) \end{array} \quad \begin{array}{c} \text{OutputEquation} \\ \left[\begin{array}{c} T_r(t) \\ T_{rr}(t) \\ RH_r(t) \\ v_{ar}(t) \\ T_{chil}(t) \end{array} \right] \\ = h(X, t) \end{array} \quad (1)$$

where the state vector is denoted by $X \in R^n$ and the outputs variables are $T_r(t)$ indoor temperature, °C, $T_{rr}(t)$ indoor radiant temperature, °C, $v_{ar}(t)$ indoor air velocity, m/s, and $RH_r(t)$ indoor relative humidity, %, $T_{chil}(t)$ child water temperature, °C, and the inputs variables are $T_o(t)$ outdoor temperature, °C, $k_2 = \frac{\sum_j A_{wj} U_j OF_b + \sum_j A_{wj} U_j OF_{rDR}}{\sum_j A_{wj} U_j OF_i + \sum_j A_{wj} h_{ij}}$ is a disturbance due to thermal resistant and solar radiation incident on the building envelope surfaces, $k_3 = \frac{\sum_j A_{wj} h_{ij}}{\sum_j A_{wj} U_j OF_i + \sum_j A_{wj} h_{ij}}$ is a convection heat transfer disturbance and thermal resistant, f_{DR} is a location effect factor, A_{slab} is a slab floor area, $f_3 = C_s \times A_L \times IDF + \dot{m}_{ven} c p_a$ (W/k) is a disturbance of internal cooling sensible heat gain caused by heat gain due infiltration and ventilation, $f_4 = f_{fen} + 136 + 2.2A_{cf} + 22N_{oc}$ (W) is the direct radiation and internal sensible heat gain caused by the occupants, lighting, and appliance for building, ω_o is the disturbance due to

humidity ratio of outdoor ($K_{g_w}/K_{g_{da}}$) and $\dot{Q}_{ig,l}$ is the latent cooling load from internal gains (W).

The relations between components of the transfer functions of subsystems for the integrated model is illustrated in Fig. 1. From the Fig. 1, it is clear the model is multiple-input parameters and multiple output variables, where the variable $\omega_r(s)$ is represented in the output vector, it is a function of time $h(X, t)$ by $RH_r(t)$.

2.2. PMV model

The Predicted Mean Vote (PMV) model can be applied as feedback control for tracking the thermal comfort level to achieve better indoor thermal comfort. The Fanger's PMV is adopted here, which is represented by a numerical indicator based on two personal variables and four physical variables. Despite the fact that this model is the closest one to the real human thermal comfort conditions but it is difficult to use in real time application due to its empirical equation is implicit type and nonlinear as summarized by Eq. (2) [30,31]:

$$\begin{aligned} PMV = & (0.303e^{-0.036M} + 0.028) [(M - W) \\ & - 3.05 \times 10^{-3} \{5733 - 6.99(M - W) - p_a\} \\ & - 0.42 \{(M - W) - 58.15\} \\ & - 1.7 \times 10^{-5} M (5867 - p_a) - 0.0014 M (34 - t_r) - 3.96 \times 10^{-8} \\ & + f_{cl} \{ (t_{cl} + 273)^4 - (t_{rr} + 273)^4 \} - f_{cl} h_c (t_{cl} - t_r)] \end{aligned} \quad (2)$$

t_{cl} , h_c , p_a and f_{cl} are given by equations:

$$\begin{aligned} t_{cl} = & 35.7 - 0.028(M - W) - 0.155 I_{cl} [3.96 \\ & * 10^{-8} f_{cl} \{ (t_{cl} + 273)^4 - (t_{rr} + 273)^4 \} + f_{cl} h_c (t_{cl} - t_r)] \\ h_c = & \begin{cases} 2.38(t_{cl} - t_r)^{0.25} & \text{for } 2.38(t_{cl} - t_r)^{0.25} > 12.1\sqrt{v_a} \\ 12.1\sqrt{v_a} & \text{for } 2.38(t_{cl} - t_r)^{0.25} < 12.1\sqrt{v_a} \end{cases} \\ p_a = & \frac{p_s RH}{100} \text{ and } p_s = \frac{C_1}{T} + C_2 + C_3 T + C_4 T^2 + C_5 T^3 + C_6 T^4 + C_7 \ln(T) \\ f_{cl} = & \begin{cases} 1.00 + 0.2 I_{cl} & \text{for } I_{cl} < 0.5 \text{ clo} \\ 1.05 + 0.1 I_{cl} & \text{for } I_{cl} > 0.5 \text{ clo} \end{cases} \end{aligned}$$

where M is the metabolism (w/m^2), PMV is the predicted mean vote, I_{cl} is the thermal resistance of clothing (m^2k/w), t_r is the room temperature (°C), W is the external work (w/m^2), t_{rr} is the room mean radiant

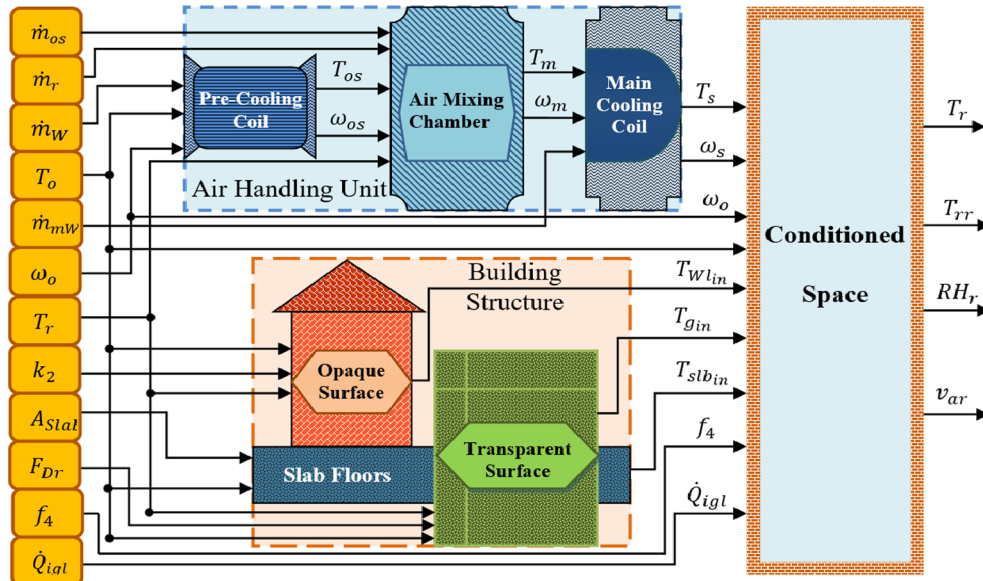


Fig. 1. Schematic diagram of integrated building and HVAC systems.

temperature ($^{\circ}\text{C}$), t_{cl} is the surface temperature of clothing ($^{\circ}\text{C}$), v_a is the relative air velocity (m/s), P_a is the water vapor pressure (Pa), f_{cl} is the ratio of the surface area of the clothed body to the surface area of the nude body, RH is the relative humidity in percentage, $C_1, C_2, \dots, C_7$ are the empirical constants that can be found from [32], P_s is the saturated vapor pressure at the specific temperature (Pa), T is the absolute dry bulb temperature ($^{\circ}\text{C}$) and h_c is the convective heat transfer coefficient ($\text{W}/\text{m}^2\text{k}$).

The Fanger's model is represented by the layer's memory framework based on Tagaki Sugeno Kang (TSK) structure, which is identified by the fuzzy logic method. This model has been developed by a previous work [32], adopted hybrid layers structure, which uses a TSK fuzzy model and tuned by Gauss-Newton method for nonlinear regression (GNMNR) algorithm to achieve an actual value of PMV. The PMV sensor model is used as the feedback gain of the indoor conditioned space to get better control performance and avoid temperature and humidity coupling, along with saving energy at the same time [33].

3. Controller design for integrated structure

The structure of the integrated two different types of fuzzy controllers consists of six outputs (regulate multiple individual compressors, solenoid valves of supplied chilled water to the pre-cooling coil and the main cooling coil, adjust dampers of the fresh and returned air and regulate the air fan speed) and four inputs for FPIPDM, error (deviation from set point), change of error, predicted mean vote (PMV) (output) and change of PMV and one input for CABTSK is a time. The control aim is to track a desired PMV profile of the output vector $[T_r(t) \ T_{rr}(t) \ RH_r(t) \ v_{ar}(t) \ T_{chil}(t)]_{DES}^T$, this requires that the output error vector $e = [T_r(t) \ T_{rr}(t) \ RH_r(t) \ v_{ar}(t) \ T_{chil}(t)]^T - [T_r(t) \ T_{rr}(t) \ RH_r(t) \ v_{ar}(t) \ T_{chil}(t)]_{DES}^T$ approaches to the vicinity of zero after a transient of acceptable (as fast as possible) duration. The main concept of integrated two different types of controllers is

illustrated by Fig. 2. The first one (FPIPDM) is solely designed to be fast as much as possible within the capacity of the driving air handling unit and regulate multiple individual compressors to ensure chilled water temperature matches demand, and the second one (CABTSK) focuses on precise human language. From Fig. 2, the system has two operation modes according to the variance value between the indoor condition and desired condition that lead to include a bang-bang mechanism (selector). In a first or default mode, the bang-bang algorithm is delegated FPIPDM, which specified for starting states or when a large variance value between the indoor condition and desired condition. In other words, this mode is active when the error between the actual value and set point value is higher than that of the threshold. When the error is below the threshold, the second mode alternate FPIPDM by CABTSK, which has better accuracy near the desired indoor condition, which is novel simple structure, as a result, computing time much reduced than for the first one.

3.1. Structure of fuzzy PI-PD Mamdani-type controller

The conventional PI-PD controller is characterized by being robust, reliable and applicable to processes that have noise, disturbances, uncertainties and large time delay [34]. The classical method PI-PD like FLC provides a means for converting a crisp PI-PD value to construct an appropriate set of membership functions (fuzzy value), this method can be found in [35]. The structural design of the FPIPDM controller is based on PI-PD like FLC, which is fairly straightforward and easy to convert a conventional controller to fuzzy controller. Therefore, the fuzzy control law is derived directly from the output of the fuzzy PI [36] and the output of the fuzzy PD [37] controller then combines them into a single final control law. The final controller uses a same set (five fuzzy sets) of MFs for defining both of its two inputs and six outputs to reduce the time of the computational. The two inputs of the fuzzy PI part are the error signal and the rate of change of error, while other two inputs of the fuzzy PD part are the average change of output and the rate of

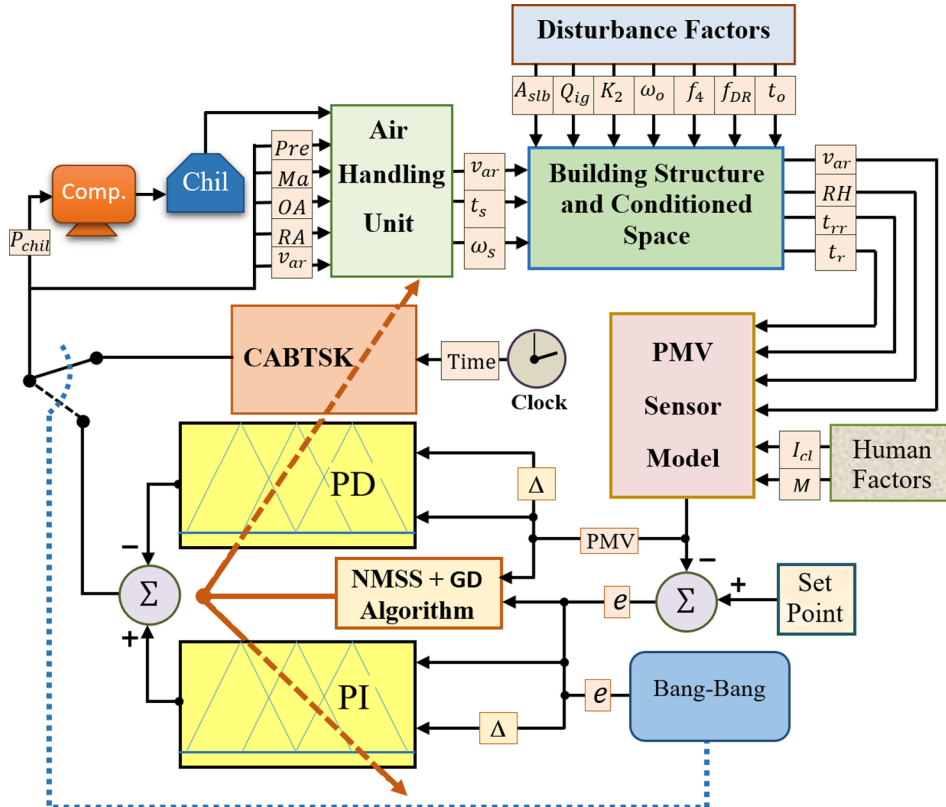


Fig. 2. The structure of integrated control of FPIPDM and CABTSK for HVAC Systems.

change of output as shown in Fig. 3. The membership function gives better result when employing triangular shape, specially, reduction time response [38]. The triangular membership functions of fuzzy sets are defined as large negative, small negative, zero, small positive, and large positive. For any fuzzy controller, it is necessary to determine the ranges of input and output variables. The universe of discourse (UOD) for the inputs is $-5 \leq X \leq 5$ and for the outputs for the main cooling coil is $-20 \leq z_1 \leq 20$, the pre-cooling coil is $-13 \leq z_2 \leq 13$, the returned air damper is $-35 \leq z_3 \leq 35$, the fresh air damper is $-35 \leq z_4 \leq 35$, the actions of the multiple individual compressors is $-6 \leq z_5 \leq 6$ and the air fan speed is $-2 \leq z_6 \leq 2$. The linguistic format based on fuzzy rules describes the relationship between the fuzzification inputs and defuzzification outputs. The operation of the defuzzification technique is one of the most effective techniques for the optimization of time response for the fuzzy controller. The largest of maxima (LOM) is fast and accurate enough for the determination of the defuzzification values than other fifth techniques (COG, FOM, MOM, MeOM, and RWA), as well as applicable methods to be employed in the optimization process [29]. However, the fuzzy PI-PD controller preserves linear behavior that leads to the use of online tuning.

The online tuning of the fuzzy logic system is a new emerging application, where its implementation of a fuzzy logic controller is limited due to the difficulty faced in tuning the fuzzy logic system [39]. In this work, the proposed controller FPIPDM is online tuned by a systematic manipulating method, which leads to a decrease time response for different initial conditions. The main objective in the online tuning of FPIPDM problem is to specify which parameter is to be optimized to get minimum response time. The design of membership function is more effective than rule base (inference engine) design on the performance of the fuzzy logic system, in particular tuning the span range of universe of discourse. In fact, the displacement of the middle span is one of the main parameters affecting time response [40]. Therefore, the spans of the middle fuzzy sets for the inputs are online manipulated from a narrow span range (S_N) to a wide span range (S_W) by an iterative process to get a suitable range of spans as shown in Fig. 4 where the center of the span is fixed. Thus, the optimization objective function $f(S)$ is related to domain or range adaptation of span for the two inputs as shown in Eq. (3). This feature is valuable at start-up time especially when high difference between the setpoint and indoor condition. The algorithm for adjustment for minimizing time response by the NMSS algorithm is described in detail in subsections 4.1 and 4.2.

$$f(S) = PMV(t, C_1^3(S_{e,er})_N^W, C_2^3(S_{PMV,PMVr})_N^W) \quad (3)$$

where t is time, $C(S)$ is a membership function for PI and PD related to spans S , subscripts e , er , PMV , $PMVr$ are error, error rate, output and output rate respectively, subscript $i = 3$ is index for optimization of the

third (middle) membership function and subscript $j = 1, 2, \dots$ is the input for two types fuzzy controller, $j = 1$ for input PI and $j = 2$ for input PD.

3.2. Structure of CABTSK

The control algorithm of cluster adaptive based on Takagi-Sugeno-Kang (CABTSK) is specified to handle a narrow margin of error. The control object is classified into clusters the represented by hybrid layers based on neural networks' weight and physical parameters' memory, which are adopted to construct a control structure. The hybrid layers are well systemized depending on the fuzzy inference system based on the Takagi-Sugeno-Kang (TSK) method. The weights of hybrid layers are primarily offline tuning for uncertainty based on evaluation cooling/heating load (clustering weight), then carry out the online fine-tuning method. Both tuning methods are nonlinear regression and conducted with the same gradient descent (GD) algorithm. The offline and online tuning methods following feedback and feedforward tuning strategies respectively. These lead to many improvements in the matter of accuracy and robustness. The structure scheme of multi-layer hybrid network controller is formulated as a four dimensions matrix (x_1, x_2, x_3, i), which is shown in Fig. 5. Fig. 5 shows that the independent input of the hybrid layers is a time variable, the primary reason for adopting this variable because the cooling/heating load as a function of outdoor conditions, which is related to time of day.

The configuration of the hybrid layers by clustering weight and parameters via break up the effect of cooling/heating load on the indoor condition profile into clusters, where the profile of indoor condition is related to cooling coil valve position. Each cluster can be represented by single basis vector of TSK fuzzy rules, then the parameters and weight layers are obtained from these rules by offline learning. Clustering technique enables the formulation of hybrid layers to extract the physical and initial neural weight parameters, and that minimize the time to perform the adaptation [41]. Furthermore, the neuron weights and physical parameters that obtained from offline learning are profound significance to reduce the tuning period, instead of its initial values as zero.

The CABTSK is classified as a multi-output control type, each of these outputs is formalized into seven groups of hyper-ellipsoid clusters as shown in Fig. 6. The clusters' weights for each output are gathered from antecedent's rule of the TSK algorithm which achieved when output of TSK fits into the cluster's curve. Hence each cluster can be represented by following nonlinear output rule.

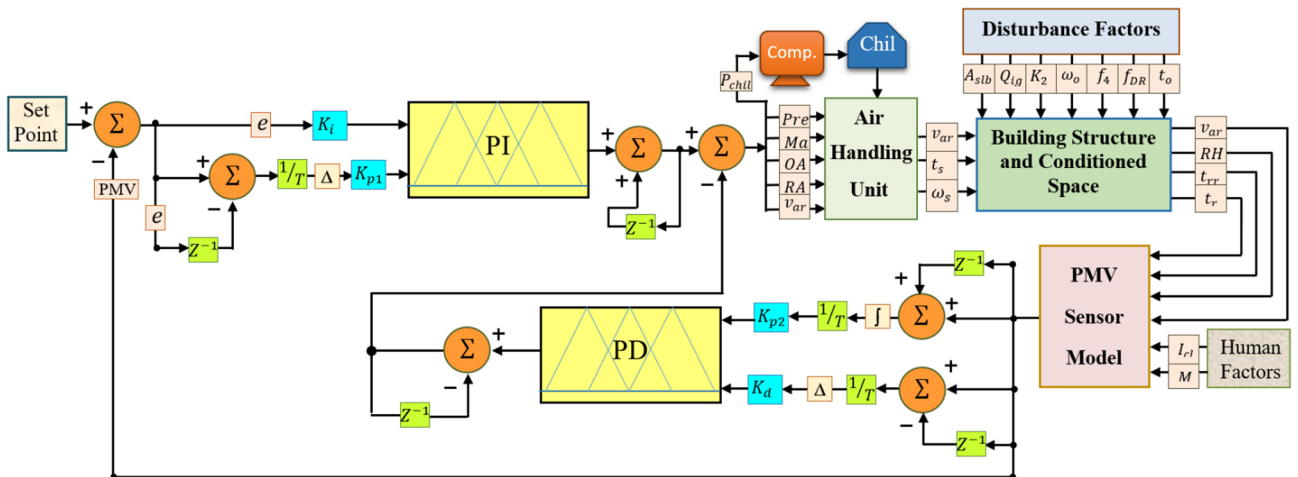


Fig. 3. Configuration of the FPIPDM structure.

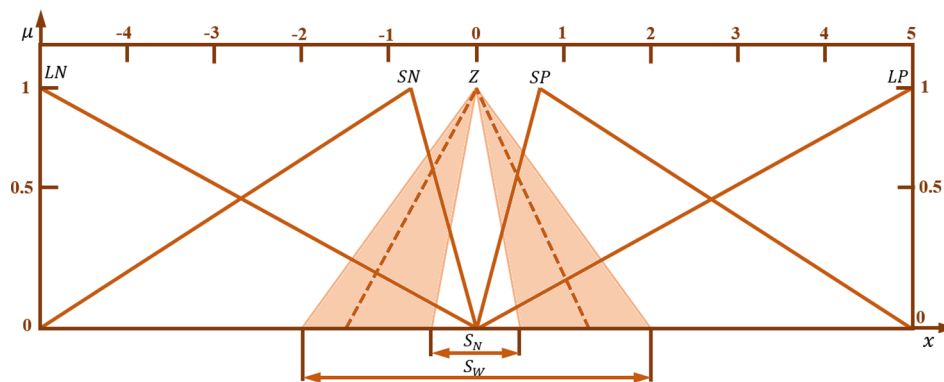


Fig. 4. Optimization of four membership functions of inputs error, change of error, MPV and change of MPV.

Cluster ($\mathfrak{R}_i$):

if x_1 is $M_i^{d(x_1)}$ and x_2 is $M_i^{d(x_2)}$... and x_m is $M_i^{d(x_m)}$ then $Y_i(X)$

$$= F_i(x_1, x_2 \cdots x_m; a_i, b_i) = \omega_i y_i \quad (4)$$

In this equation, the independent input parameters vector of UOD ($X = [x_1, x_2 \dots x_m]^T$) at antecedent rule part is the time of the day, which is the most significant effect on the changes in output in the consequent part of the rule. The cluster rule is formulated in a general form that where it can be seen the parameters and weight are correlated to the time which is the input of the model, the m is represented the number of input parameters. To fit the output of TSK into the cluster's curve, the linguistic values $d(x_1), \dots, d(x_m)$ are needed to be described well. Whereas the value part of a cluster rule is related to the nonlinear output $F_i(x_1, x_2 \dots x_m; a_i, b_i)$, it is a function to parameters (a_i and b_i) and weights (ω_i).

The premise (input) data set of each controller output can be partitioned into clusters, each centre point of a cluster is correlated basis functions and number of $\mu A(x)$ (membership functions) as explained earlier in Fig. 6. Then inference (combining of antecedent(s)/input(s)) and aggregating the output of clusters are implemented to obtain conjunctive/disjunctive modelling output as follows:

Cluster($\mathfrak{R}_i$):

if x_1 is $M_i^{d(x_1)}$ and x_2 is $M_i^{d(x_2)}$... and x_m is $M_i^{d(x_m)}$ then $Y(X)$

$$= \sum_j Y_i(X) = \sum_{i=1}^N \omega_i y_i \quad (5)$$

In the consequence part, the cluster number is represented by a subscript j and it seems to be interpreted a singleton type. The center of gravity (COG) defuzzification technique is one of the most common techniques used for fuzzy mathematics defuzzification method to get the crisp value of each cluster. The discrete version can be expressed in the following formula:

$$Y(X) = \frac{\sum_{i=1}^N \beta_i(u) y_i}{\sum_{i=1}^N \beta_i(u)} \quad (6)$$

It is obvious from Eq. (6) that we can obtain the cluster output of the i^{th} rule that corresponds to the activation degree $\beta_i(u)$, using N fuzzy linguistic term set. The accumulated activation degree of all the rules can be represented as follows:

$$\begin{aligned} \beta_i(u) &= \mu M_i^{d(x_1)}(x_1) \wedge \mu M_i^{d(x_2)}(x_2) \wedge \cdots \wedge \mu M_i^{d(x_m)}(x_m), \\ 1 \leq i \leq N \end{aligned} \quad (7)$$

According to the modified Eq. (6), which is a singleton output, it is shown that it gets more accurate of the weight value to achieve better

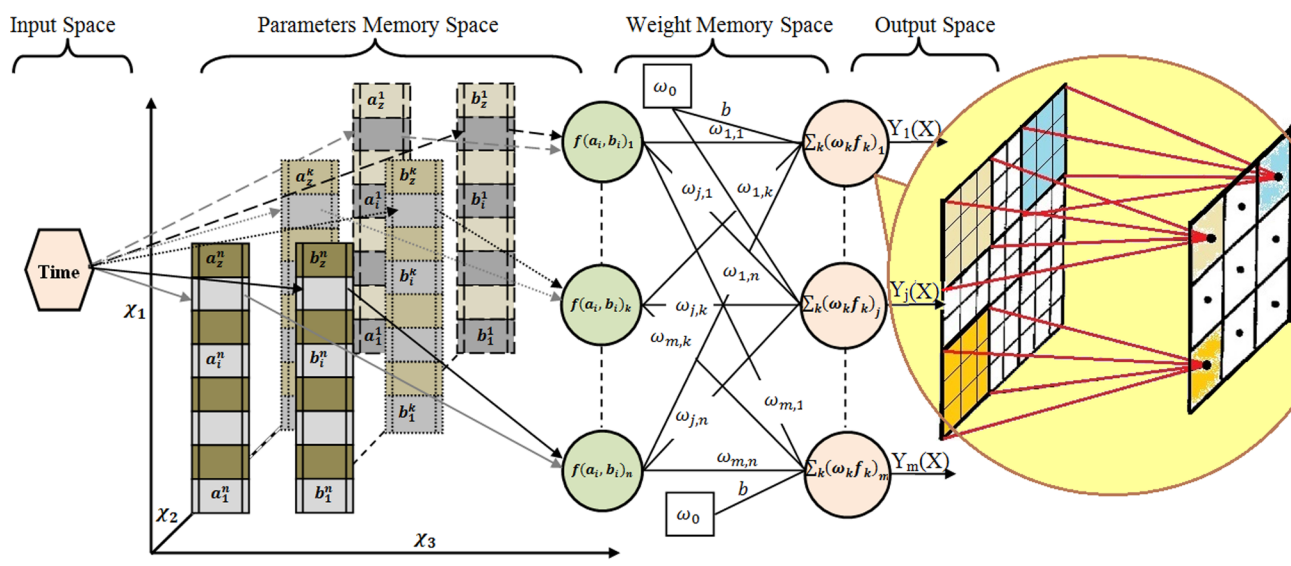


Fig. 5. Schematic design of multi-layer hybrid of CABTSK controller.

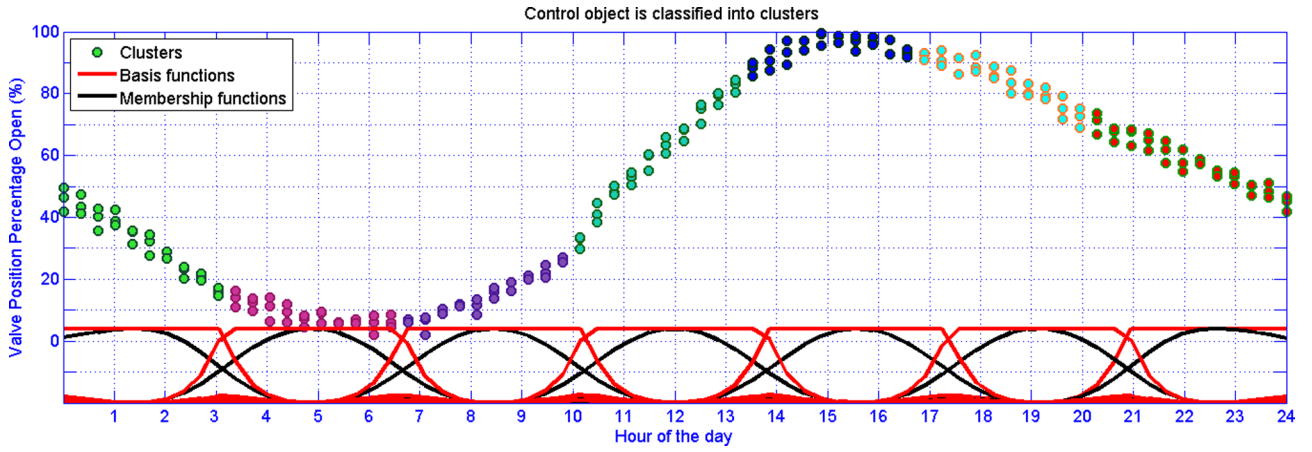


Fig. 6. Clustering data set of main cooling coil valve position based on premise membership's functions and basis.

fitness for clustering consequences. [41]. The final equation form can be specified as follows:

$$Y(X) = \frac{\sum_{i=1}^N \beta_i(u) y_i}{\sum_{i=1}^N \beta_i(u)} = \sum_{i=1}^N \omega_i y_i \quad \omega_i = \frac{\beta_i(u)}{\sum_{i=1}^N \beta_i(u)} \quad \beta_i(u) = \prod_{j=1}^m \beta_{ij}(u_j) \quad (8)$$

The correlation of nonlinear part of the singleton output in Eq. (6) is expressed in exponential form as follows:

$$Y(X) = \sum_{i=1}^N \omega_i a_i (1 - e^{-b_i x}) \quad (9)$$

The weights (ω_i) and parameters (a_i and b_i) in Eq. (9) can be manipulated by adjusting them from the input–output relations in the consequent part at Eq. (4). The singleton coefficients (parameters and weights) are classified as two types of the layer according to its location and sequence in Eq. (9), to store in memory space. The coefficients are systemised into relevant groups of hybrid layers which highly correlated with time to be the input of the control model, which is shown in previous Fig. 5. The first groups of layers are trained by cooling load data in order to produce parameters layers, whereas the basis function of the clustering is used to identify weights layers of the neural network.

Since the task of the CABTSK is to track the set point precisely and to be more accurate than FPIPD, hence the optimization algorithms are used to minimize errors due to uncertainty of coefficients of hybrid layers (the control model). The optimization function in the GD algorithm is well-suited to handling process constraints without decomposing them into setpoints as described in detail in the next subsection.

3.3. Pre-tuned offline of CABTSK

The offline optimization of hybrid layers is carried out by convert cooling load profile into seven groups of clusters according to the piecewise outputs of TSK, which can be expressed as:

Cluster ($\mathcal{R}_i$):

$$\text{if } x_1 \text{ is } M_i^{d(x_1)} \text{ and } x_2 \text{ is } M_i^{d(x_2)} \dots \text{ and } x_m \text{ is } M_i^{d(x_m)} \text{ then } Y(X) = \sum_{i=1}^N \omega_i a_i (1 - e^{-b_i x}) \quad (10)$$

The hybrid function $Y(X)$ is divided into small parabolic based on cluster centre to align it with the cooling load, these segments are optimized by GD algorithm through adjusting value of parameters ω_i , a_i and b_i in Eq. (10). The offline tuning techniques based on nonlinear

regression to obtain the parameters' value are implemented by using the GD iteration to provide tight convergence to the cooling load profile. The error mechanism is adopted by GD learning, the following equations can be formed using the expression error.

$$y_i = \omega_i f(x_i; a, b) + e_i \quad (11)$$

The measured value of the dependent variable is represented by y_i , whereas the independent input variables $\omega_i f(x_i; a, b)$ are sorted into two groups, each with different attributes and types varies across time, such as ω_i , a_i and b_i which has a non-linear relationship (exponential) with the dependent variable and affected by the independent variable x_i . The independent input variables are significantly affected by residual measurement error e_i . The cluster centre point is used in a polynomial approximation to a multivariate Taylor series expansion regarding three variables (ω_i , a , b) to represent hybrid layers. The value of the variation derivative can be predicted by partial derivative as follows:

$$f(x_i)_{q+1} = f(x_i)_q + \frac{\partial f(x_i)_q}{\partial a} \Delta a + \frac{\partial f(x_i)_q}{\partial b} \Delta b + \frac{\partial f(x_i)_q}{\partial \omega} \Delta \omega \quad (12)$$

The correction of the three variables in the offline optimization process, takes to implicate error on the final equation by subtracted Eq. (12) from Eq. (11), yielding the following Equation.

$$y_i - f(x_i)_q = \frac{\partial f(x_i)_q}{\partial a} \Delta a + \frac{\partial f(x_i)_q}{\partial b} \Delta b + \frac{\partial f(x_i)_q}{\partial \omega} \Delta \omega + e_i \quad (13)$$

The partial derivatives can be represented by using a matrix scheme to contain all difference of the Jacobian matrix $[Z_q]$ as follows:

$$\{D\} = [Z_q] \{\Delta A\} + \{E\} \quad (14)$$

The dependent variable $\{D\}^T = [d_1 \ d_2 \ \dots \ d_n]$ of the difference between the real and function values is used to obtain the tuning column vectors in $\{\Delta A\}$ define as unknown coefficients $\{\Delta A\}^T = [\Delta a_0 \ \Delta a_1 \ \dots \ \Delta a_m]$ which $\Delta a = a_{q+1} - a_q$, $\Delta \omega = \omega_{q+1} - \omega_q$ and $\Delta b = b_{q+1} - b_q$. The vector of residual error $\{E\}^T = [e_1 \ e_2 \ \dots \ e_n]$ is eliminated through the iterations.

The inverse of a matrix $[Z_q]^{-1}$ can be used for obtaining parameters' vector $\{\Delta A\}$ in Eq. (15) by implementing the GD algorithm technique.

$$\Delta A = [[Z_q]^T [Z_q]]^{-1} \{[Z_q]^T \{D\}\} \quad (15)$$

Finally, the initial value of parameters' vector $\{\Delta A\}$ of unknown coefficients is estimated by using the iterative numerical method. The offline tuning is a viable strategy for reducing the execution time of online tuning as demonstrated in subsection 4.3.

4. Online tuning

The minimum response time can be achieved by tuning the span of membership functions. The relation between the indoor PMV and spans of the FPIPDM controller membership functions are optimized automatically to achieve minimum response time by using NMSS method based on heuristic search. The optimization is an iterative process by searching the parameters of span S in FPIPDM controller.

4.1. NMSS algorithm

The Nelder-Mead simplex search (NMSS) algorithm distinguishes in simple, derivative-free iterative (not to be confused with numerical or analytical gradients), multi-objective optimization and fully programmable at a hardware level. The simplex dimensions determined by set of points which is the convex hull of its vertices, where the n -simplex is defined by subset of the $n + 1$ point ($f(C_{n+1}), C \in R^n$).

The outline of the iterative steps of NMSS algorithm can be summarized as the following: the objective function's values of the simplex vertices are ordered ascending then find out reflection point (C_r) by calculating the centroid of all points except(C_{n+1}) such that.

$C_r = C_c + \alpha \times (C_c - C_{n+1})$, C_c is represented the centroid of points in the simplex vertices and $\alpha > 0$

In case of the reflected point, it is better than all points except the best one (C_1), then the new simplex is obtained by replacing the reflected point (C_r) with the worst one (C_{n+1}) and repeats the above series actions. Then it obtains the new expanded point by $C_e = C_c + \beta \times (C_r - C_c)$ where $\beta > 1$. When the value of expanded point (C_e) is better than the reflected (C_r) that means a new simplex can be obtained by replacing the worst point (C_{n+1}) with the expanded one (C_e) then repeat the above series actions. Otherwise (C_r) is better than (C_e), the new simplex is obtained by replacing the worst point (C_{n+1}) with the reflected point (C_r) from there repeating the above series actions. If the reflected point is the worst point ($f(C_r) \geq f(C_n)$) so, the new point is a contracted point such that ($C_{Co} = C_c + \rho \times (C_{n+1} - C_c)$) with $0 < \rho \leq 0.5$. If $f(C_{Co}) < f(C_{n+1})$ then the contracted point is better than the worst point, at this status the new simplex can be obtained by replacing the worst point (C_{n+1}) with the contracted point (C_{Co}) then repeats the previous series actions. To evaluate the objective function values at the end of the iteration when convergence is reached to ($f(C_{Co}) > f(C_{n+1})$) then replace all simplex points except the best (C_1) with ($C_i = C_1 + \sigma(C_i - C_1)$).

4.2. Online tuning of the FPIPDM

The penalty function method for nonlinear constrained optimization is adopted for online tuning in NMSS algorithm. The NMSS algorithm is implemented by passing the four parameters of the input's membership functions (S_e , S_{er} , S_{PMV} , S_{PMVr}) into the NMSS algorithm searches for the optimal value of the objective function by recursively

computing. The ends of S (same span value) are preferred to identical to the peak values of inputs MFs SN and SP. As can be seen from Fig. 4, the best setting for the initial value for each parameter of S to $2.5(S_e = S_{er} = S_{PMV} = S_{PMVr} = 2.5)$, to achieve 50% overlap to fulfill requirements for Demaya's repartition [42]. The distribution of MF is followed by the Bezdek distribution to prevent the modal points of the MF from crossing each other. The initial guess values for the span has so far reached an acceptable quality standard.

Then, the optimization process is adopted based on the penalty function method with S parameters to work against deviationism from the desired indoor condition (error of PMV). The main objective of the NMSS algorithm is to minimize the penalty function value. The desired time response is achieved when getting an effective penalty function, which is difficult to find [43]. From the objective function in Eq. (3) the global minimizer of the quadratic penalty function can be found explicitly as follows:

$$g_{\min} = \sum_i^N (f_i(S))^2 \quad (16)$$

By applying Eq. (16) as a penalty function that leads to minimize the time response, such act is accompanied by overshoot due to nonlinear penalty function. The oscillations or overshoot take place by reversing output under the effects of the NMSS algorithm has taken output signal to negative search. Hence the oscillations are damped out by modifying penalty function which can be mathematically described as in Eq. (17):

$$g_{\min} = \sum_i^N (|f_i(S)|)^2 \quad (17)$$

The application of the penalty function Eq. (17) to tune the parameters of the premise part yields positive PMV and minimum response time.

4.3. Post-tuned online of CABTSK

In defiance of FPIPD scheme it permits a more rapid response to an initial condition, especially when high difference between setpoint and indoor condition but the optimization response takes a long time since three processes involved in the FPIPD; fuzzification of inputs, an inference engine or a rule base, and a crisp output obtained from defuzzification in addition, it has two loops. While the response of optimization of the CABTSK is faster, that yields from converting the cluster of TSK inference system into hybrid layers. This feature is valuable to skip the logical operation task and it is time-consuming.

The online optimization of hybrid layers is carried out by starting from the initial value of parameters' vector $\{A\}$, which is obtained by offline optimization. Since the initial value is dramatically drifted due to uncertainty of outdoor/indoor disturbances, it is so crucial to perform online tuning. The GD algorithm is often used to handle the optimization for multi-parameters, this makes CABTSK easier to manage

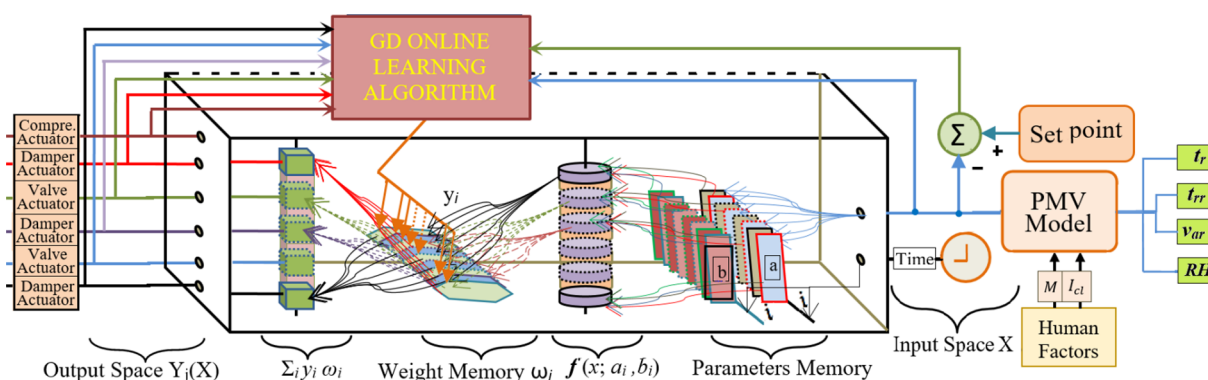


Fig. 7. Online tuning schedule of GD for the CABTSK.

intricate characteristics that emerge from varying outdoor/indoor conditions, it is fully utilised for online tuning, as depicted in Fig. 7.

Since the GD is the lower number of tuning iterations than the LMB and the GD is based on steepest descent greedy search the value of the parameters' vector $\{A\}$ from the global non-stationary points. That converge the value of error $\{E\}^T = [e_1 \ e_2 \ \dots \ e_{n+1}]$ becomes very small at a minimum value of the objective function $F(A_n)$. The subscript n is used to denote dimension of the column vector of the unknown coefficients, $(A_1, A_2, \dots, A_{n+1})$. Accordingly, the structure of hybrid layers needs a direct search algorithm for non-linear optimization and that is exists on the NMSS algorithm. Hence, the values of the parameters vector are tuned based on the plant output deviation from set point to reduce the error.

The GD algorithm is to seek for the ΔS of direction for maximum rate (the steepest-ascent trajectory) of the gradient (the partial derivative) of the hybrid layers model outputs $\partial Y(t(q); \omega_i, a_i, b_i)$. The formula for vector tuning rule can be expressed by the following equation.

$$A_i(q+1) = A_i(q) + \Delta S_i(q) \cdot \frac{\partial Y(t(q); \omega_i, a_i, b_i)}{\partial A_i(q)} \quad (i = 1, \dots, n) \quad (18)$$

The variable of frequent daily time t is the input argument of hybrid layers' structure at the number of tuning iterations q . the clustering uncertainty misrepresents the hybrid plant model outputs that lead error is associated with the predicted output $Y(t(q); \omega_i, a_i, b_i)$. Therefore, the parameters (arguments of the weight and constant) $[\omega_i a_i b_i]^T$ requires tuning until decaying error over time. The variance evaluation of the gradient error at iteration scheme is given by.

$$\nabla e_{q+1} = \left[\frac{\partial}{\partial \omega_i} Y(t_{q+1}; \omega_i, a_i, b_i) \quad \frac{\partial}{\partial a_i} Y(t_{q+1}; \omega_i, a_i, b_i) \quad \frac{\partial}{\partial b_i} Y(t_{q+1}; \omega_i, a_i, b_i) \right]^T \quad (19)$$

Equation (19) needs to be iterated to decay error in order to obtain a more accurate value for the hybrid layers' parameters and weights of the $[\omega_i a_i b_i]^T$ vector.

$$\begin{bmatrix} \omega_i \\ a_i \\ b \end{bmatrix}_{q+1} = \begin{bmatrix} \omega_i \\ a_i \\ b \end{bmatrix}_q + \begin{bmatrix} \Delta \omega_i \\ \Delta a_i \\ \Delta b \end{bmatrix}_q \quad (20)$$

The updated value of parameters and weights $\begin{bmatrix} \Delta \omega_i \\ \Delta a_i \\ \Delta b \end{bmatrix}_q$ is done by getting the gradient of error ∇e_{q+1} , which can be presented in the following formula.

$$\begin{bmatrix} \Delta \omega_i \\ \Delta a_i \\ \Delta b \end{bmatrix}_q = \Delta S_q \nabla e_{q+1} \quad (21)$$

In each iteration used optimal step ΔS size is used, which is an objective function of gradient descent search. The gradient deviations of the output of the hybrid layers lead to errors, it can be described by the following equation:

$$e_{q+1} = e_q + \Delta e = mpv(t_{q+1}) - S \cdot P_q = \sum_{i=1}^N \omega_i f(t_{q+1}; a_i, b_i) - S \cdot P_q = Y(t_{q+1}; \omega_i, a_i, b_i) - S \cdot P_q \quad (22)$$

The current and desired (setpoint) indoor conditions are represented respectively by $mpv(t_{q+1})$ and $S \cdot P_q$ at iteration $q+1$ and q .

The number of clusters characterized by N which is descended from linguistic term set.

The nonlinear Eq. (20) is continuously updated feedback, this type can be introduced only in the expansion of multivariable for Taylor

formula to tune the values of the weights and parameters, as described by the following first derivative equation:

$$e_{q+1} = e_q + \Delta e = Y(t_q; \omega_i, a_i, b_i) + \frac{\partial}{\partial \omega_i} Y(t_q; \omega_i, a_i, b_i) \Delta \omega_i + \frac{\partial}{\partial a_i} Y(t_q; \omega_i, a_i, b_i) \Delta a_i + \frac{\partial}{\partial b_i} Y(t_q; \omega_i, a_i, b_i) \Delta b_i - S \cdot P_q \quad (23)$$

$$e_{q+1} = e_q + \frac{\partial}{\partial \omega_i} Y(t_q; \omega_i, a_i, b_i) \Delta \omega_i + \frac{\partial}{\partial a_i} Y(t_q; \omega_i, a_i, b_i) \Delta a_i + \frac{\partial}{\partial b_i} Y(t_q; \omega_i, a_i, b_i) \Delta b_i \quad (24)$$

Then, continuing iteration as long as the error changes to get the set of sequence of the ΔS (step-length), in order to form optimal path of the tangent to the graph of the error equation, and thus by substituting the values of the parameters and weights vector $[\Delta \omega_i \Delta a_i \Delta b_i]^T$ of Eq. (23) into Eq. (24). The recursive Equation reaches a terminating condition by setting the error equal to zero, the following steps will be followed to achieve the algorithm goal.

$$e_{q+1} = e_q + \nabla \left(\frac{\partial}{\partial \omega_i} Y(t_q; \omega_i, a_i, b_i) \Delta S_q + \frac{\partial}{\partial a_i} Y(t_q; \omega_i, a_i, b_i) \Delta S_q + \frac{\partial}{\partial b_i} Y(t_q; \omega_i, a_i, b_i) \Delta S_q \right) \quad (25)$$

$$e_{q+1} = e_q + \frac{\partial^2}{\partial \omega_i^2} Y(t_q; \omega_i, a_i, b_i) \Delta S_q + \frac{\partial^2}{\partial a_i^2} Y(t_q; \omega_i, a_i, b_i) \Delta S_q + \frac{\partial^2}{\partial b_i^2} Y(t_q; \omega_i, a_i, b_i) \Delta S_q \quad (26)$$

$$e_q + \frac{\partial^2}{\partial \omega_i^2} Y(t_q; \omega_i, a_i, b_i) \Delta S_q + \frac{\partial^2}{\partial a_i^2} Y(t_q; \omega_i, a_i, b_i) \Delta S_q + \frac{\partial^2}{\partial b_i^2} Y(t_q; \omega_i, a_i, b_i) \Delta S_q = 0 \quad (27)$$

Finally, the following Eq. (28) is reached to obtain the value of ΔS_q by simplifying Eq. (27).

$$\Delta S_q = \frac{-e_q}{\frac{\partial^2}{\partial \omega_i^2} Y(t_q; \omega_i, a_i, b_i) + \frac{\partial^2}{\partial a_i^2} Y(t_q; \omega_i, a_i, b_i) + \frac{\partial^2}{\partial b_i^2} Y(t_q; \omega_i, a_i, b_i)} = \frac{S \cdot P_q - \omega_i(q) a_i(q) (1 - e^{-b_i(q) t_i})}{\omega_i(q) a_i(q) e^{-b_i(q) t_i}} = \frac{S \cdot P_q - \omega_i(q) a_i(q)}{\omega_i(q) a_i(q) e^{-b_i(q) t_i}} + 1 \quad (28)$$

From offline tuning estimate the initial values of weight and parameters, then applying these values in Eq. (28) to get the initial value of ΔS_0 . So as to initiate the use online optimization technique to find the optimal values of weight and parameters is implemented by Eq. (20), which becomes as follows:

$$\begin{bmatrix} \omega_i \\ a_i \\ b \end{bmatrix}_{q+1} = \begin{bmatrix} \omega_i \\ a_i \\ b \end{bmatrix}_q + \left(\frac{S \cdot P_q - \omega_i(q) a_i(q)}{\omega_i(q) a_i(q) e^{-b_i(q) t_i}} + 1 \right) \nabla e_{q+1} \quad (29)$$

From Eq. (29), it is relatively straightforward technique to optimize weights and parameters of hybrid layers based on gradient of error or its rate of change, which is used as feedback for the online tuning.

5. Physical description for the bang-bang and evaluation indices

5.1. Bang–bang (on/off)

In this paper, the bang-bang (on/off) is subjected to do two tasks, as a benchmark comparative and to select right mode under specified error threshold value. After the indoor condition reaches error threshold value, the bang-bang controller switches from FPIPD control (mode 1) to TSK control (mode 2). Its deadband operating fraction can be described by the following equation.

$$\rho = \frac{t_{md1}}{t_{md1} + t_{md2}} \quad (30)$$

where

$$t_{md1} = \frac{1}{r} \ln \frac{PMV_{set} - \frac{1}{2} PMV_{hys} - PMV_{md1}}{PMV_{set} + \frac{1}{2} PMV_{hys} - PMV_{md1}} \text{ and } t_{md2} = \frac{1}{r} \ln \frac{PMV_{set} - \frac{1}{2} PMV_{hys} - PMV_{md2}}{PMV_{set} + \frac{1}{2} PMV_{hys} - PMV_{md2}}$$

In these equations, r is the indoor PMV decay rate time constant, PMV_{set} is the PMV set-point, PMV_{hys} is the bang-bang's hysteresis, PMV_{md1} is the steady-state PMV when the mode 1 is on, and PMV_{md2} is the steady-state PMV when the mode 2 is on. This duty cycle corresponds to the probability that we observe a device to be on at any given time.

5.2. Performance indices

In addition to the indoor thermal comfort level is used as an overall performance of controllers, the energy consumption is inevitable to compare controllers of three different structures. The overall power consumption can be described by the following equation [44].

$$I_e = \int_{t=0}^{24} (P_c(t) + P_h(t) + P_f(t)) dt \quad (31)$$

where, P_c is the cooling power, P_h is the heating power and P_f is the fan power

The implementation of the Eq. (31) is required to obtain the control volume energy balance of each device in the HVAC systems (cooling coil, heating coil, and air supply fan). This study adopted the control volume of the pre-cooling coil, air supply fan and the main cooling coil to assess the cooling load for an HVAC system. The cooling load via the pre-cooling coil and the main cooling coil is calculated based on water-side inlet and outlet parameters which are the chilled water flow rate and its temperature. The chilled water flow rate is related to the control signals of the opening position of the chilled water valves as described in detail in the subsections 6.3 and 7.3. Equation (31) can be represented as follows:

$$P_c(t) = \dot{m}_c(t) cp (T_{out}(t) - T_c(t)), \quad P_h(t) = \dot{m}_h(t) cp (T_h(t) - T_{out}(t)) \text{ and } P_f(t) = \alpha \dot{m}^3(t)$$

In these equations, cp is the specific heat capacity of water = 4.186 (kJ/kg.°C) and α is the fan power constant = 0.5 (kW.s³/kg³)

5.3. The statistical indices

The criteria of statistical indices are clearly performance assessment of the hybrid layers model, the most significant standards of statistical indices are the Maximum absolute error (*Max. AE*), the Mean squared error (*Mean SE*) and the Mean Absolute error (*Mean AE*), which can be defined as follows:

- Maximum absolute error;

$$\text{Max. AE} = \max |y_i - y_{p_i}| \quad (32)$$

- Mean absolute error;

$$\text{Mean AE} = \frac{1}{N} \sum_{i=1}^N |y_i - y_{p_i}| \quad (33)$$

- Mean squared error;

$$\text{Mean SE} = \sqrt{\frac{1}{N-1} \sum_{i=1}^N |y_i - y_{p_i}|^2} \quad (34)$$

where y_i is the calculated or real values, y_{p_i} is a simulated values and N is the numbers of testing samples.

6. The validation and evaluation

This section presents an overview of the validation building and PMV models to run and evaluate the proposed controller in a complicated state similar to real-time. Thus, it needs identification of the building's physical characteristics.

6.1. The building and PMV models

Interconnected control and models to the evaluated system are required to introduce models' variables and parameters for transfer function models of Equations (1 and 2). A simple structure building has a single-story floor with the air handling unit devices are adopted in this study. Building's gross ground floor area and the interior and exterior wall height are 248.6 m² and 3.5 m respectively, whereas the net building area without garage area (total conditioned space area) is 200 m², and it has one master room of three bedrooms, one living room, two bathrooms, utility, kitchen, two little storage, and open garage. 25.08 m² of external walls are face north, 25.6 m² face south, 47.1 m² face east and 5.2 m² face west side. The walls' thickness is 18 cm and its material are cement, bricks, concrete, EPS (Expanded Polystyrene), plaster and foam insulation. The windows overall area is 1.44 m² face south, 5.13 m² face north, and 7.29 m² face east, for more details of building description can be found in Table 1.

The indoor building model response is shown in Fig. 9 depending on outdoor measured condition of real spring season day, where the spring dry bulb temperature oscillated from (18–33 °C) to (22–39 °C), relative humidity from (68–98%) to (35–70%), predicted mean vote (PMV) from (-0.18–1.8) to (0.1–2.6) and predicted percentage of dissatisfaction (PPD) from (1–82) to (5–96), where the indoor condition has much less latitude (limits of the upper and lower band) than outdoor due to the building's insulation. This occurs when the indoor dry bulb temperature oscillated from 23.4 °C to 37.3 °C, relative humidity from 41% to 70%, PMV from 0.24 to 2.25, and PPD from 5 to 89. The modeling

Table 1
The properties of the materials that are construction.

Component	Quantity	Variables	Notes
Ceiling	200 m ²	U = 0.18 W/ (m ² . K) Light α_{roof} = 0.4	Overall area less courtyard area (22.6 × 11) – (7.3 × 7.3) 2 (each 0.9 by 2.1 m)
Doors	3.8 m ²	U = 2.3 W/ (m ² . K)	
Windows	13.9 m ²	Fixed: U = 2.84 W/ (m ² . K); SHGC = 0.67 Operable: U = 2.87 W/ (m ² . K); SHGC = 0.57 T _x = 0.64; IAC _{cl} = 0.6	
Walls, exposed exterior	126.2 m ² gross, 108.5 m ² net	U = 0.7 W/ (m ² . K)	Wall height = 2.4 m
Walls, garage	35.0 m ²	U = 0.7 W/ (m ² . K)	Include perimeter adjacent to the courtyard Wall gross area (including courtyard wall) plus ceiling area
Floor area	95.3 m ²	R _{cw} = 0.21 W/ (m ² . K)	
Floor perimeter	67.2 m		
Total exposed surface	67.2 m ²		
Volume	356.5 m ³		

validations, implemented with real measurement data and calculated values of output parameters. The building model validation was carried out under the end-of spring conditions, on the building in Basra city, as depicted in Fig. 8. The 24 - hour outdoor conditions are inputs into the model, the model outputs (temperatures and RH), and indoor measurements data are simultaneously recorded. The measurement of indoor temperature and RH has been taken at each time step (at a minimum interval of 15 min.). The comparison between the simulated and measured results showed slight differences between them. Therefore, the simulated results agree with the measured results according to statistical indices resulting from Fig. 9. The statistical results of the model outputs indicate that each one got a significant correlation, $r^2 = 0.9982$ for temperature and $r^2 = 0.8998$ for RH as shown in Table 2. Whereas, for the model of Fanger (PMV, PPD) verification, it was subjected to the same conditions as those imposed upon the building model. Thus, to prove the validity of the Fanger model, its output results are compared with numerical calculations, which are based on Eq. (2). The calculation was implemented every 20 min. for 24 h by considering the building model outputs as its input parameters. It can be seen obviously from Fig. 10 that the calculated results are smaller than the simulation results, due to the fact that the model was driven by TSK structure so that the nonlinear equation was represented well. Therefore, according to Table 2, the correlation coefficient showed that there was a relationship between the model outputs and the numerical calculations, where $r^2 = 0.9997$ for PMV and $r^2 = 0.8705$ for PPD.

6.2. The FPIPDM performance analysis

The PD drives the inner feedback loop to support the dominated cascaded PI controller in the outer loop. The three-dimensional control action surface for each part of the control shows its burden of contribution, therefore the control surface or action surface clarifies control performance. The surface shape of the PD part is mountainous, such visualization action surface is extremely valuable and aggressive to be more active and faster transitions, as shown in Fig. 11. This surface has a stationary gain near the center of the output (PMV) and output rate (PMVr) plane, and elsewhere the gain is rapidly increased. Whereas the nonlinear control surface of the PI part is incongruent with the PD part, it also has big deviation with the linear surface, as shown in Fig. 12. This kind of visualization, exhibits that PI has a higher output gain value than the linear surface plane at the center of the error signal and

the rate of error. It is obvious that a combination of the PD in the inner feedback loop with PI in the primary or outer feedforward loop provides much better performance such transient response than traditional, by taking advantage of the LOM technique to yield controller that has an optimum time response (fast rise time).

6.3. Validation of the hybrid layers

From the subsection 3.2 structure of CABTSK and Fig. 5, it is obvious that the hybrid layers of CABTSK has one effective input that is related to cooling load which is a sample time and six outputs (regulate multiple individual compressors, solenoid valves of supplied chilled water to the pre-cooling coil and the main cooling coil, adjust dampers of the fresh and returned air and regulate the air fan speed) related to damper/valve position signal (% open). The criteria of statistical indices are used for comparative evaluation of simulation and real numerical calculations for each output. For instance, when looked closely at one of the most effective action steps is to influence the outputs of the indoor cooling load, which is the main cooling coil valve position. The performance validation of the hybrid layers of CABTSK for a valve position of main cooling coil is tested by comparing it with the numerical calculations. The numerical calculations of valve position depend on the energy balance for the main cooling coil and the CLF/CLTD_c (cooling load factor for glass/corrected cooling load temperature difference) technique [26]. The comparison results of simulated and calculated of valve position and its statistical indices are shown in Fig. 13, where it can be observed that the offline tuning of parameters of the hybrid layers by GD algorithm yield considerable performance. Here, the statistical indices of mean square error, maximum absolute error and mean absolute error between the calculated valve position and the valve position obtained from the hybrid layers of CABTSK model were 0.1070, 0.2514 and 0.09078, respectively.

7. Results and discussion

The scope of this paper is to present the control performance analysis of the proposed technique by comparing it against two benchmark reference controllers, the PID and bang-bang (ON/OFF) controllers.

7.1. Evaluation of control response to indoor PMV

There are a lot of indices that are used to express the human thermal



Fig. 8. Illustration of the tested building and schematic plan showing details of the spaces and fixtures.

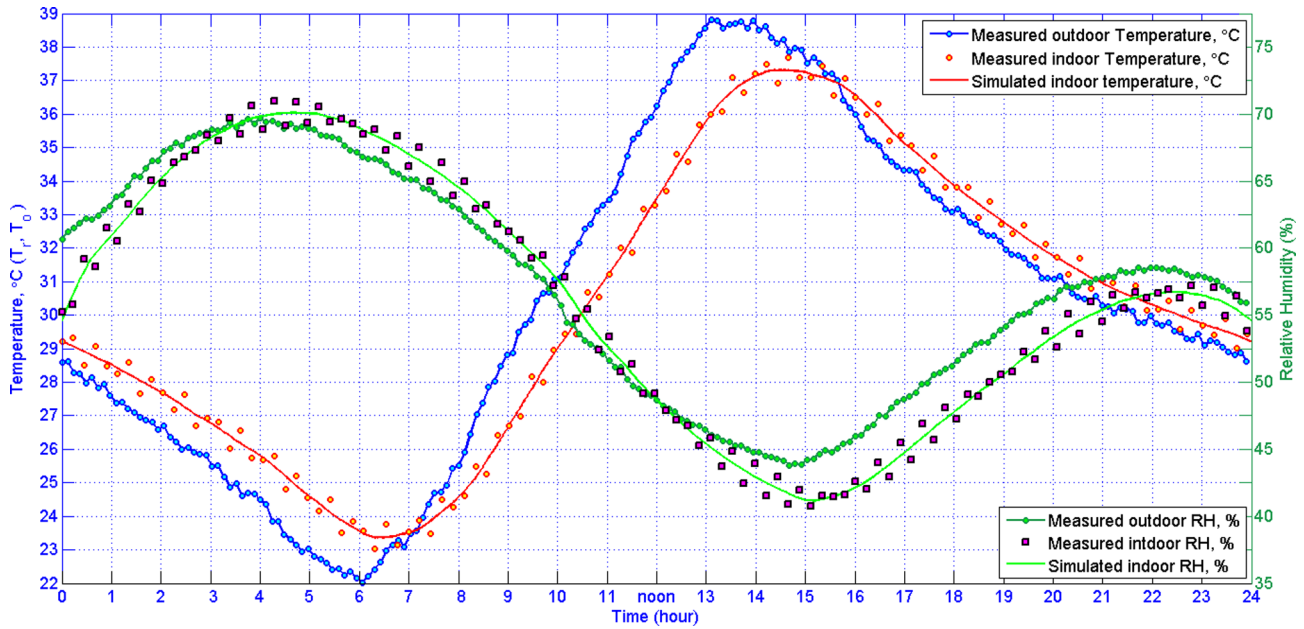


Fig. 9. Comparison between the measured outdoor data and the indoor response data by identified model to temperature and relative humidity.

Table 2

Validation of modeling using statistical indices with measured and calculated values of output parameters.

Output parameters	Performance index					Units
	Max. AE	Mean SE	Mean AE	r^2	Cov. %	
Temperature	0.3657	0.0221	0.0495	0.9982	2.206	°C
Relative Humidity	0.00687	0.002977	0.003848	0.8998	23.062	%
PMV	0.04790	0.01971	0.03532	0.9997	1.879	–
PPD	0.9390	0.1177	0.2996	0.8705	12.731	%

comfort, the closest one to the real behavior of the thermal sensation is PMV [45]; therefore, it is suggested to be a reference of controllers to conduct indoor desired conditions. The main objective of the proposed controller is to get minimum response time for the desired objective

variable (PMV), along with much reduced energy usage. From Fig. 9, it is noticeable that the indoor temperature gradient is larger at 11:30, therefore it is taken as starting point to run the three HVAC systems. The starting point at 11:30 means the highest accelerative point of indoor gaining load, which needs the best efforts of the HVAC system against the rising gaining load. After running the three different controllers' scheme simultaneously, it is definitely their performance varying substantially as indicated in Fig. 14. From Fig. 14, it is clearly the online tuning of FPIPD influence on the six model inputs to get minimum response time to achieve one of the most important objectives. Thus, it is reasonable when the numerical results reveal a distinct contrast in the required time to reach the desired objective (the set point of PMV) for each of the three controllers. To demonstrate this fact, look closely at the indoor behavior from the point of inhabitants' interpretations of their thermal comfort level as shown in Fig. 15. From Fig. 15, it is obvious that the proposed controller was the fastest one

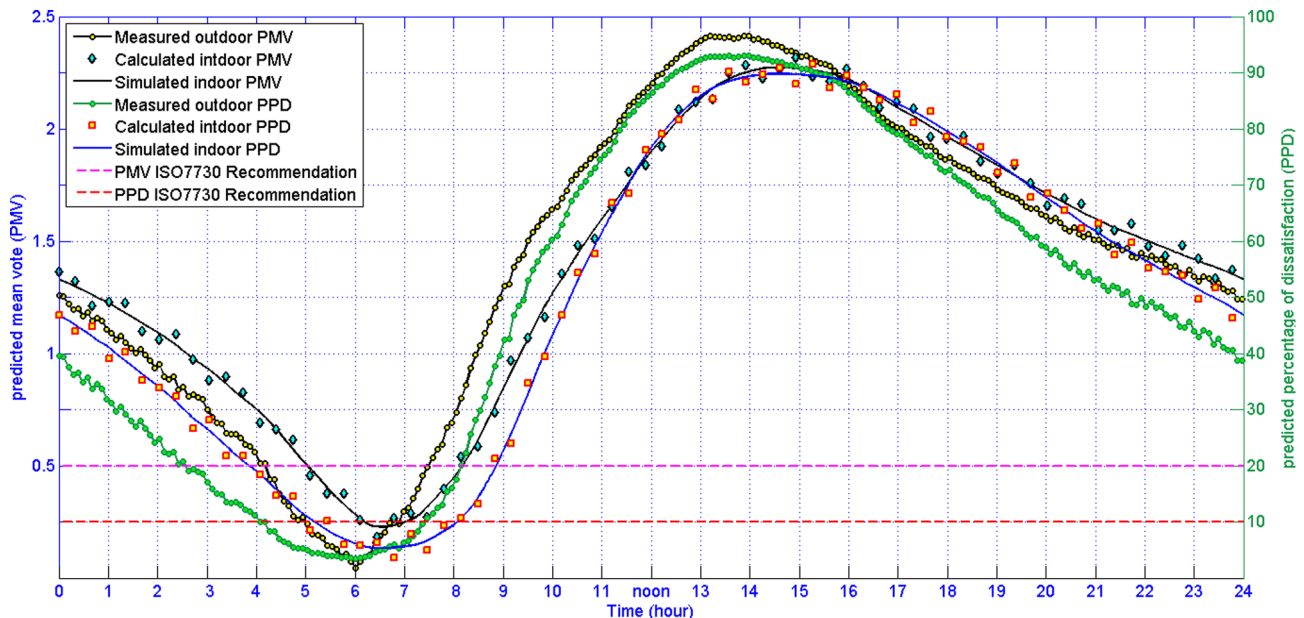


Fig. 10. Comparison between the measured outdoor data and the indoor response data by identified model to PMV and PPD.

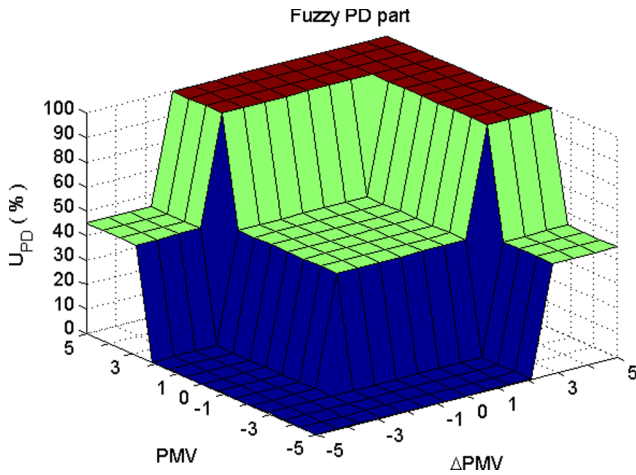


Fig. 11. The visualization of surface plots of Fuzzy PD gain parameters.

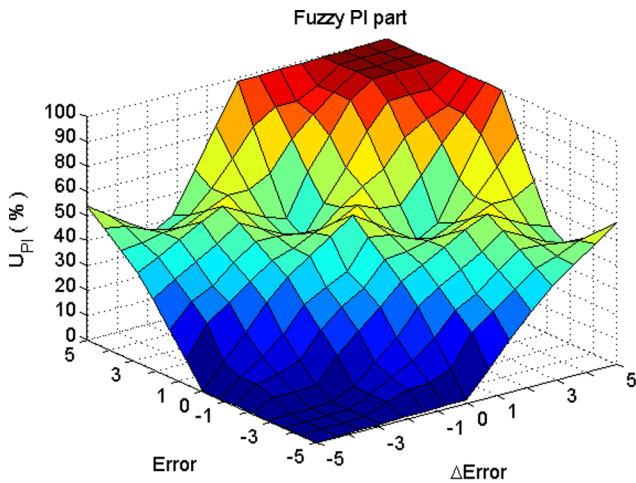


Fig. 12. The visualization of surface plots of Fuzzy PI gain parameters.

that required 4 min. and 18 sec. to reach the set-point, while the traditional PID required 13 min. and 7 sec. and the On/Off required 13 min. and 15 sec. It is easy to distinguish that the two benchmark controllers (PID and On/Off) have taken more than three times longer than the proposed controller. This shows the ability of the first part of the proposed controller which has efficiently utilized the manipulated variables offered by AHU, besides, handling the adjustment operation of the multiple individual compressors to meet the cooling load requirements. The second mode is to delegate control tasks to the second part of the proposed controller which is designed based on the hybrid layers of CABTSK, it is formulated according to four dimensions structure. The CABTSK is appointed to minimize the steady-state error of the desired indoor conditions. So, the second aim of the study is achieved when selecting the CABTSK by feedback signal at a specified error threshold value to provide a high level of output accuracy as shown in Fig. 15, it is obviously there is a significant variation in error between the three controllers. The mean absolute error for the traditional PID, On/Off and proposed FPIPDM are 0.3971, 0.3548 and 0.1316 respectively, and these values indicate that the online tuning of the CABTSK is extremely diligent in carrying out. In conclusion, there is a clear distinction between acts of the three controllers, the two benchmark controllers exhibited an inferior act in their performances and that was reflected in violated standard's lines of the recommended indoor thermal comfort, as evident in Fig. 14. On the contrary, the proposed controller passed through recommended indoor thermal comfort on to its superiors of performance. In regard to the two benchmark controllers' violation, it is evident by observing indoor temperature act as explained in the next subsection.

7.2. Evaluation of control response to indoor temperature and RH

The most variable effect on the PMV is the temperature as can be deduced from equation (2); therefore, the trend of temperature and PMV curves are substantially the same as how it can be things, but there is a slight difference in behavior, such as those shown in Figs. 14 and 16. The suggested design of the online optimization algorithm makes the proposed controller use its best efforts to keep the temperature at the standard of ISO 7730 recommended for the comfortable environment as shown in Fig. 17. To further confirm the reliability and keep

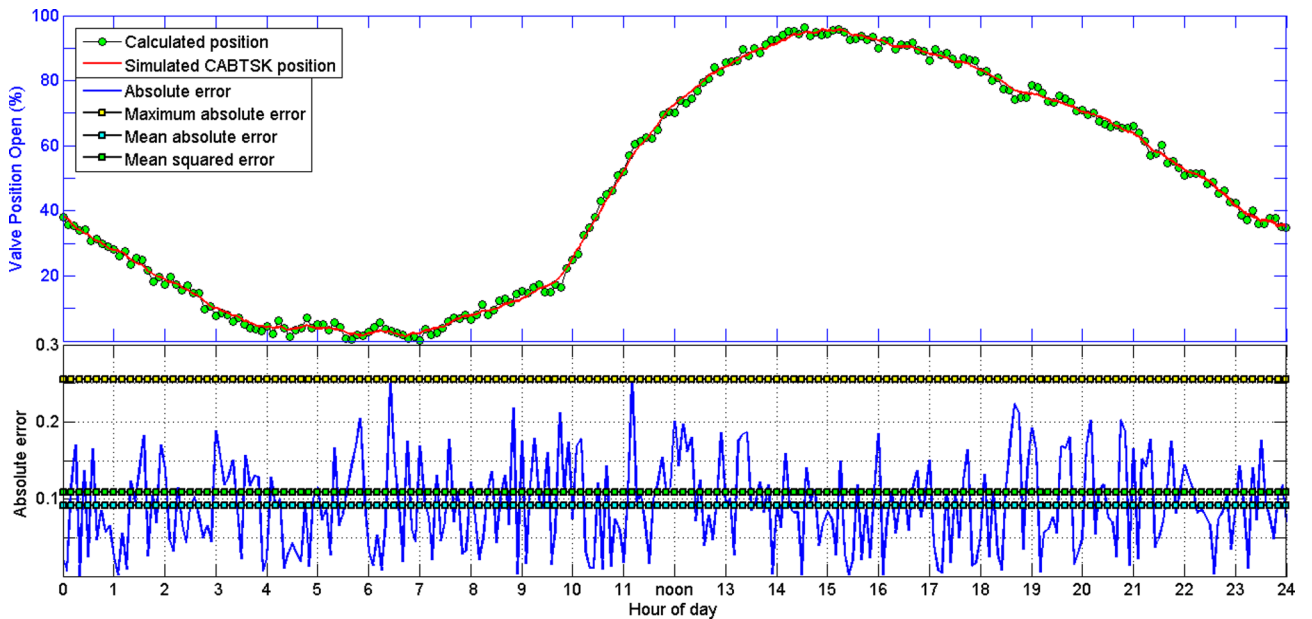


Fig. 13. Statistical indices to judge performance of CABTSK model for main cooling coil valve position.

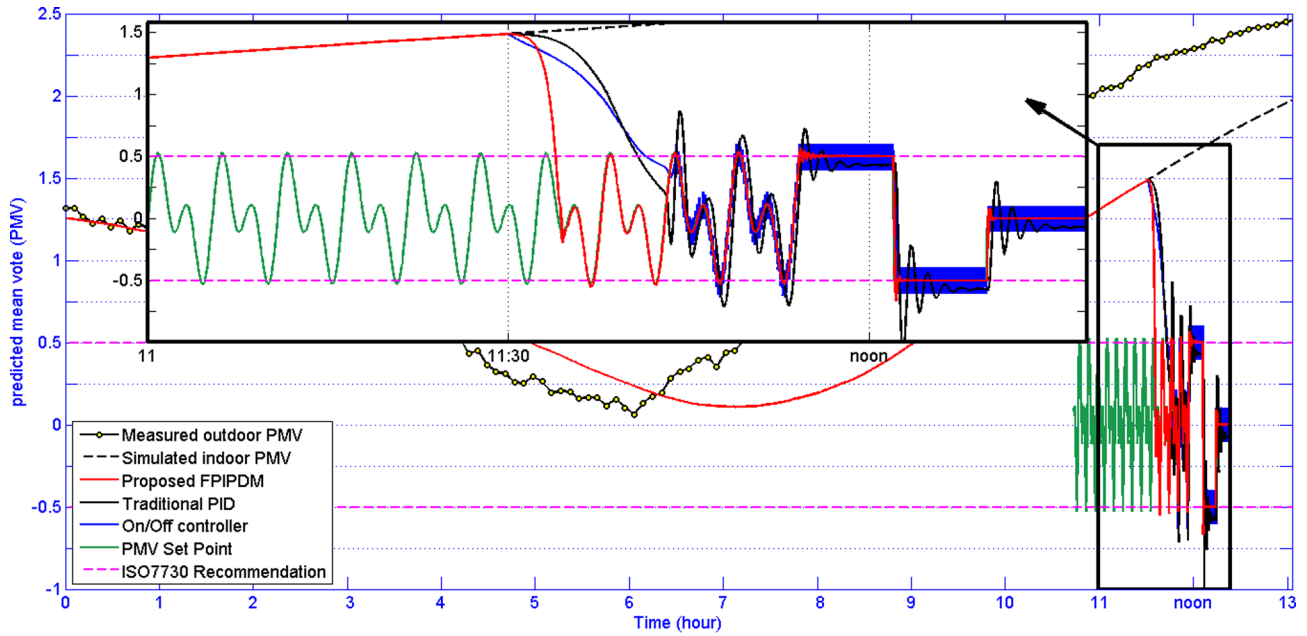


Fig. 14. Shows starting point, target location and ending point of three controllers to judge performance of proposed controller.

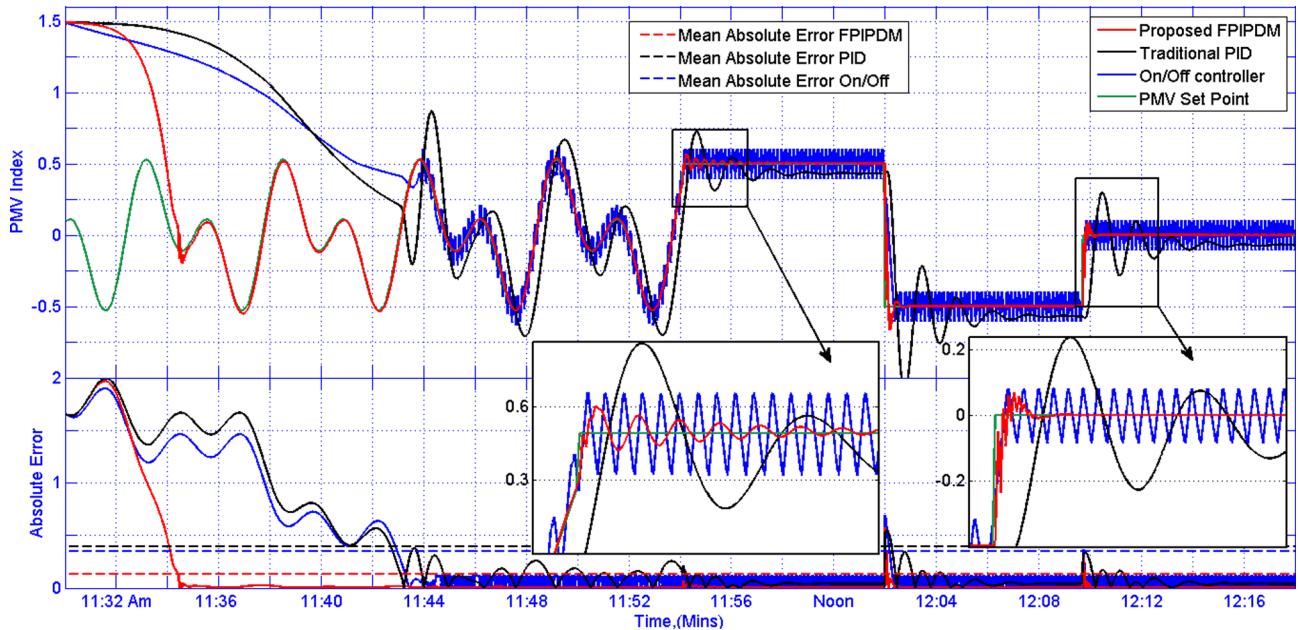


Fig. 15. Illustrate a comparison between the three controllers and shows statistical indices to judge the performance of the proposed controller.

occupants comfortable, here is another criterion. According to the recommended indoor temperature ranges based on (ASHRAE standard 55, [26]) the standard for acceptable operative temperature ranges, the proposed controller has verified good satisfaction, unlike others benchmark, they violating the ASHRAE standard many times as can be seen in Fig. 18.

Commonly, the preferred relative humidity level varies from 30 to 55 percent [41], and the RH is the least variable effect on the PMV as can be expected based on equation (2), which makes it challenging to handle and to guide in the right conditions. Although hard to control, it is inducted within recommended limits by the proposed controller, but it fluctuated when conducted by other controllers, where the worst-case fluctuation when conducted by On/Off controller and the worst-case indoor conditions when conducted by PID controller, as shown in Fig. 17.

7.3. Evaluation of control response to indoor PPD and valve position

The Predicted Percentage of Dissatisfied (PPD) index is an index that represented a quantitative of thermal sensation dissatisfied for persons who feel too warm or too cool, and its numeric value ranging from 0 to 100. The PPD is closely related to the PMV index due to both indices predict the thermal comfort and their mathematical concepts are related to each other. To evaluate the PPD index behavior for three controllers are implemented simultaneously in the presence of outdoor/indoor disturbance and modeling uncertainty, the quantitative and qualitative result showed that the proposed controller is overcoming in the uncertainties and disturbances. This is evident in Fig. 19, which shows the proposed controller conduct indoor environment into recommended condition within 4 min. and 9 sec. without violation of the recommended area due to exploiting the online tuning to track the

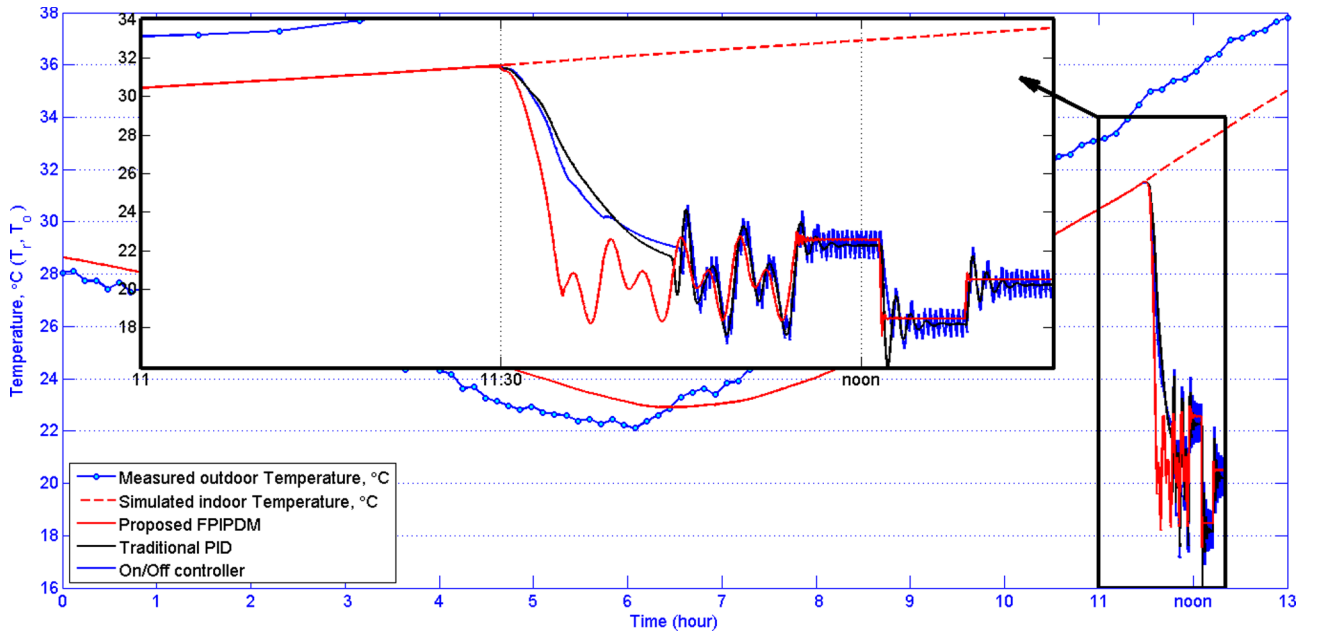


Fig. 16. Assessments of tradeoffs between the three controllers to judge performance of proposed controller.

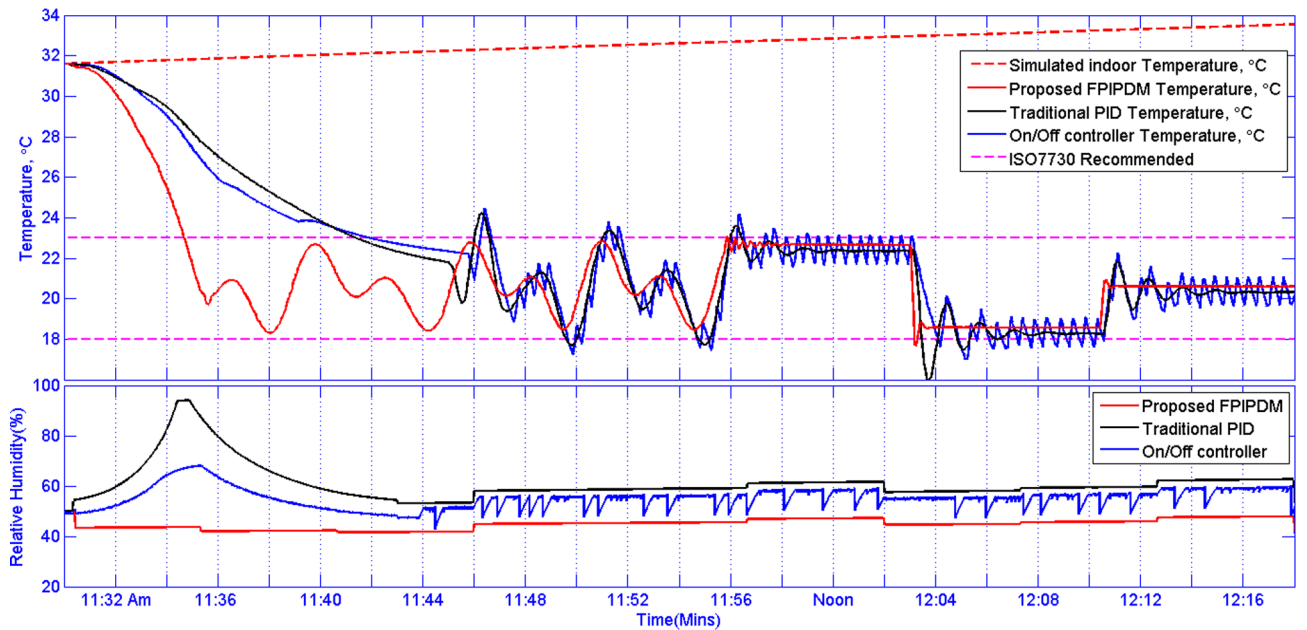


Fig. 17. The comparisons of indoor temperature between three the controllers and RH to judge performance of proposed controller.

response of set-point signal (PMV) significantly as shown in Fig. 15. While the other two benchmark controllers when employed spent much time to get a recommended condition (more than 11 min. and 34 sec.). Furthermore, they are characterized as an oscillator's response and that lead to the crossover of the recommended area frequently. This oscillation was caused by an unstable valve position, which can be seen in Fig. 19, this mechanism specified as a notorious feature due to cause severe wear in the actuators for valves, dampers, and compressors. These occur only with an inefficient controller and its reaction to any change that requires a long time to get a settlement which is contrary to the proposed controller.

7.4. Analysis of psychrometric processes

The thermodynamic and physical properties of moist air can be well

described dynamically by a psychrometric chart that leads to obtaining occupants' perceptions related to levels of satisfaction. Furthermore, the psychrometric chart is the most advanced feature available in climate assessment and how this influences human thermal comfort. According to ASHRAE standard 55–92 [46], the region of the thermal comfort zone on the psychrometric chart is indicated by contour lines. By knowing any two properties (such as temperature and RH) of air-conditioned space, the state of moist air can be represented on the psychrometric chart. Therefore, the dynamic of indoor environmental conditions can be followed and understanding the process of air conditions change with time throughout the running of HVAC systems. Thus, when starting running the HVAC systems in cooling mode, the psychrometric chart is very essential for a better description of indoor conditions for assessing the performances of three controllers. This is where these controllers are subject to the same demands in terms of a

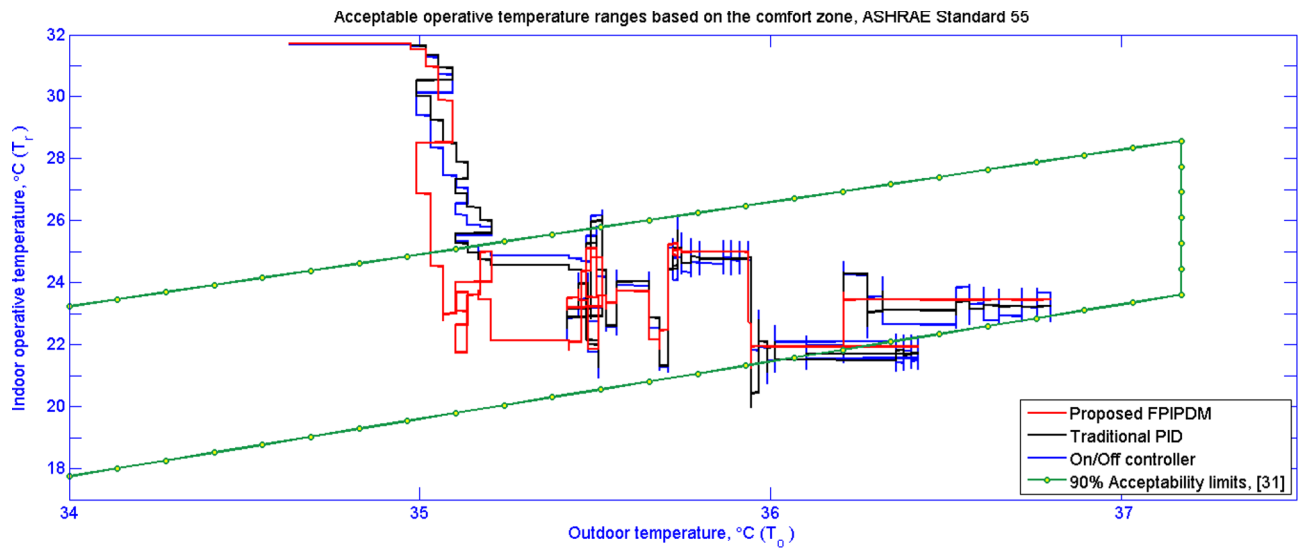


Fig. 18. Proposed controller assessments according to ASHRAE standard 55 of acceptable operative temperature ranges of.

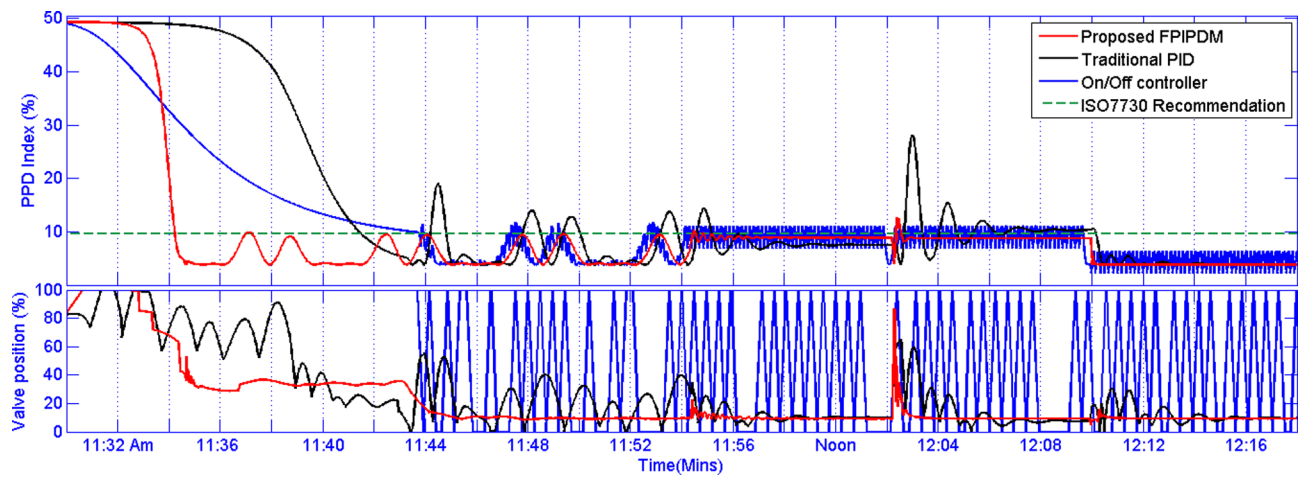


Fig. 19. Evaluate the PPD index behave and effect of valve position for three the controllers.

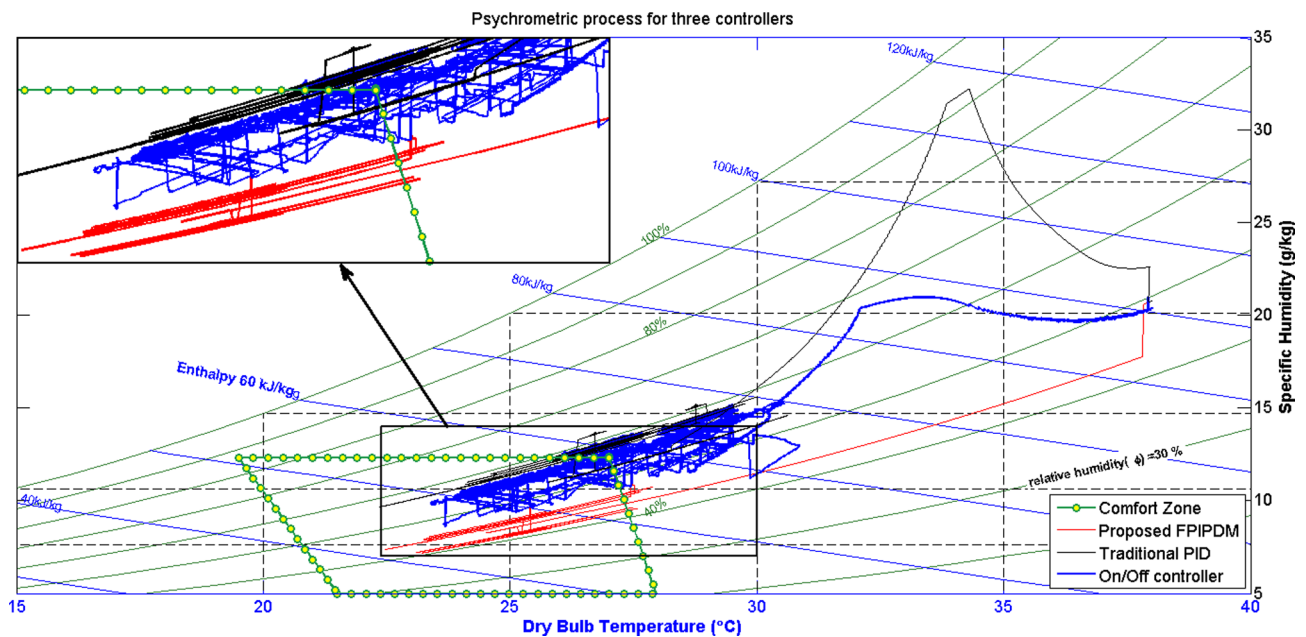


Fig. 20. Evaluate the three controllers behave via comfort zone of psychrometric chart.

significant contribution to control the quality of the indoor conditions. The abilities and performances of the three controllers are clearly different from each other as shown in Fig. 20 over 48 min. All the three controllers crossed boundary region of comfort zone, but in varying ways and in varying degree of overlaying. Fig. 20 shows the lower degree of overlaying a comfort zone area is On/Off controller and the larger degree of overlaying is a proposed controller which is almost full degree within a comfort zone area. This demonstrates the best performance of the proposed controller is suggested to conduct HVAC systems due to combined two types of the controller a faster one is an aggressive (FPIPDM) and a robust one is an accurate response (CABTSK).

7.5. Comparing energy analysis results

As can be seen from the previous subsections, the proposed controller improves the HVAC system performance, whereas the two benchmark controllers are caused by a serious violation of standards of indoor conditions. The ever-increasing time of overshoots and undershoots of the indoor thermal comfort will ensure increased losses in energy efficiency on the HVAC system [41]. This causes an additional power requirement for chillers, pumps, fans, etc. due to multiple causes, such as the nonlinearity in thermodynamics and the circulating power at the time of running HVAC systems, and controller set-point errors, that have a substantial influence by control structure design. The calculation for the energy consumption of an HVAC system needs to define which parts of each element are most related to the control signal of the cooling load. In fact, the cooling coil takes up almost the entire energy supplied to the HVAC system. It is evident that the valve position is a function of the indoor cooling load, on the other hand, the control chilled water valve signals are represented the input signals to the main and pre-cooling coil as shown in Fig. 21. The behavior of various energy consumption for a cooling load of a residential building is obtained as a result of computational modeling as explained in Fig. 22. The numerical techniques for modeling are based on Eq. (31) that are used to solve partial differential equations such as building energy consumption and cooling loads. Based on Fig. 22, the average path for each cooling coil load for the three different systems proposed FPIPDM, traditional PID and On/Off controller are 8.91 kW, 11.04 kW and 13.78 kW respectively. By running a cumulative sum on the Eq. (31) for three different types of controllers to get their power consumption trend which varies among three controllers as shown in Fig. 23. The proposed control strategy shows a reduction of power consumption to get a marked improvement in performance, which achieved extremely high levels of energy-saving around 37.05% 24 h in a day of the On/Off controller. The great variation of the increment trend of the power curve for the

three controllers is shown in Fig. 23; the recession trend in the proposed controller curve is indicated as improving in their performance, the recorded values of three types proposed FPIPDM, traditional PID and On/Off controller are 192 KWh a day, 237 KWh a day and 305 KWh a day respectively.

In the case of a constant ($PMV = 0$) set-point, it is certainly the duty of the three systems that is much less than used set time-varying between -0.5 and 0.5 , due to the fact that the system almost under steady-state conditions. In other words, in this case, eliminating the effect of transient time response, which is evident in Figs. 22 and 23.

8. The real time applications

One of the advantages that the proposed controller (FPIPDM + CABTSK) offers is a reduction technique for a real-time implementation of computational cost. This is achieved because the two parts of controller requires a smaller number of iterations to perform the learning/training and tuning procedure, which is carried out using the NMSS and GD algorithms. In addition to this, due to the implementation in real-time, the error gained in the simulations need not be this rigorous and therefore, will further time to adding more flexibility to the bang-bang code delegation state as well reduce the tuning time. This is one of the possible options for the errors, the number of iterations will be reduced by half if the error criteria are brought up from 0.128 for the mean square error, 0.286 for the maximum absolute error, and 0.154 for the mean absolute error to 0.261, 0.382 and 0.307, respectively. This increased error is considered acceptable when compared with two benchmark controllers (traditional PID and On/Off controller) that lead to reduced iteration by 50%. As regards the number of iterations, the number of iterations is based on the fitness of each cluster, the identification of the cluster required to be fitted according to the seven basis, the fourth cluster (centroid of the clusters) requires 12 iterations for tuning its parameters, the third and fifth takes 10 iterations for each cluster, and the remaining clusters take 8 iterations for each one, the total iterations are 64. Therefore, controller tuned to follow set-point changes tends to respond fast, the tuning iterations are executed for each sampling time or time step, the optimizer of the model inputs is used to measure the execution time of a loop iterations in each sampling. According to Ramakrishnan [32] analysis based on a popular microcontroller type M16C/62P, the sampling time for the 64 iterations is 1.2 Sec (each iteration takes about 20 mSec). This is much less than what was expected and then what Castilla [32] found, where they found the sampling time was 5 min.

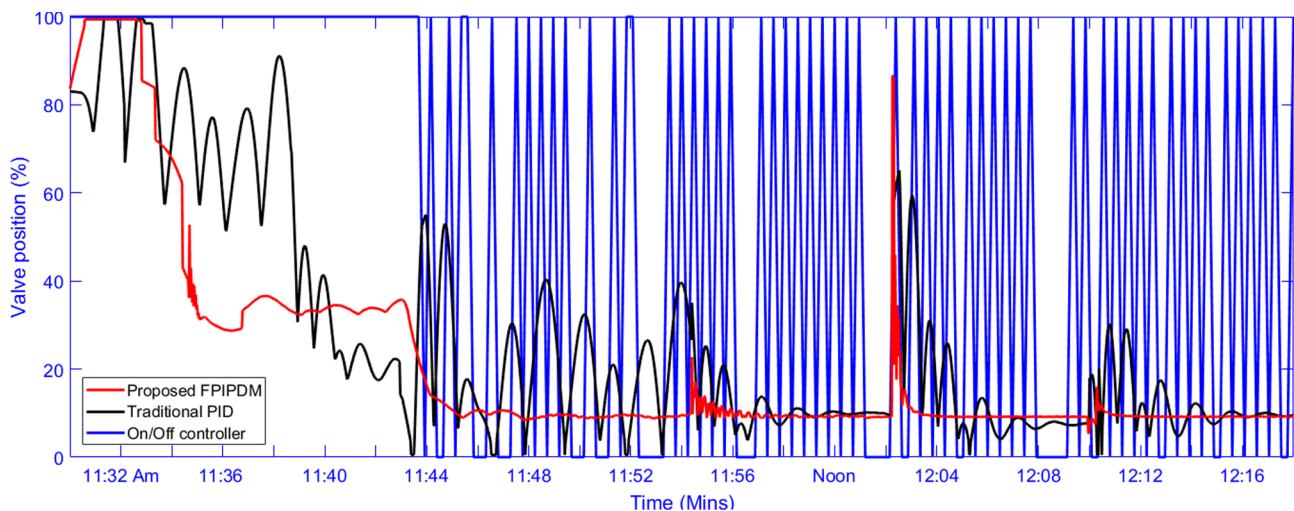


Fig. 21. The HVAC system main cooling coil valve position signal (open %) of the three controllers.

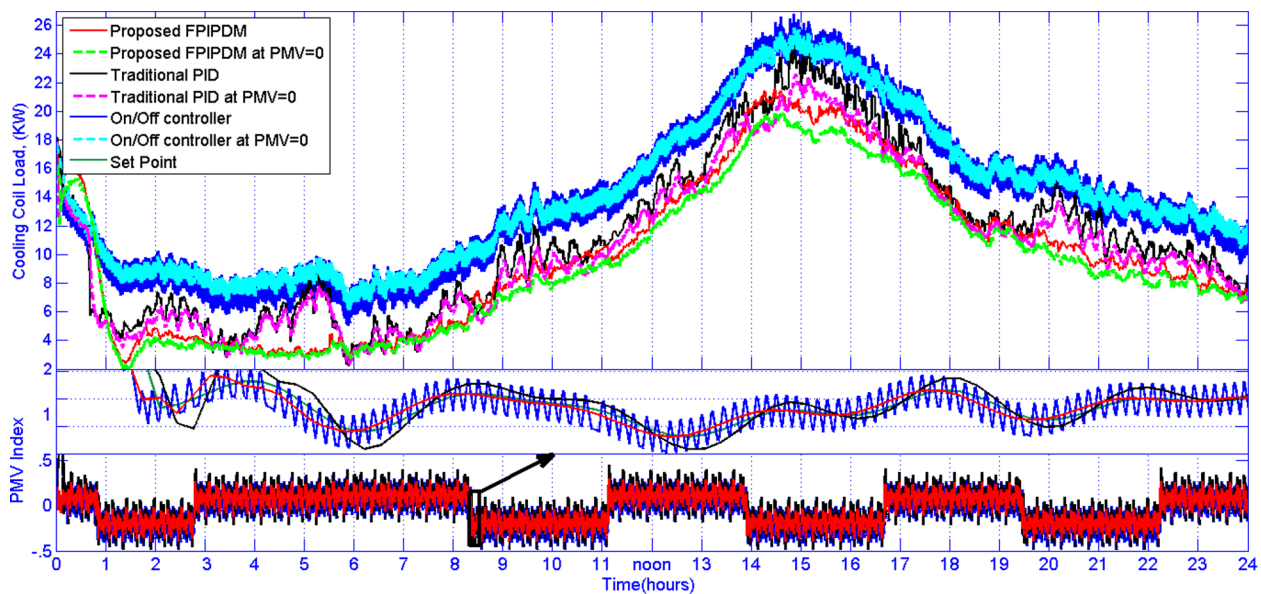


Fig. 22. Differences among the three controllers' scheme by calculated cooling coil load energy consumption of the HVAC systems.

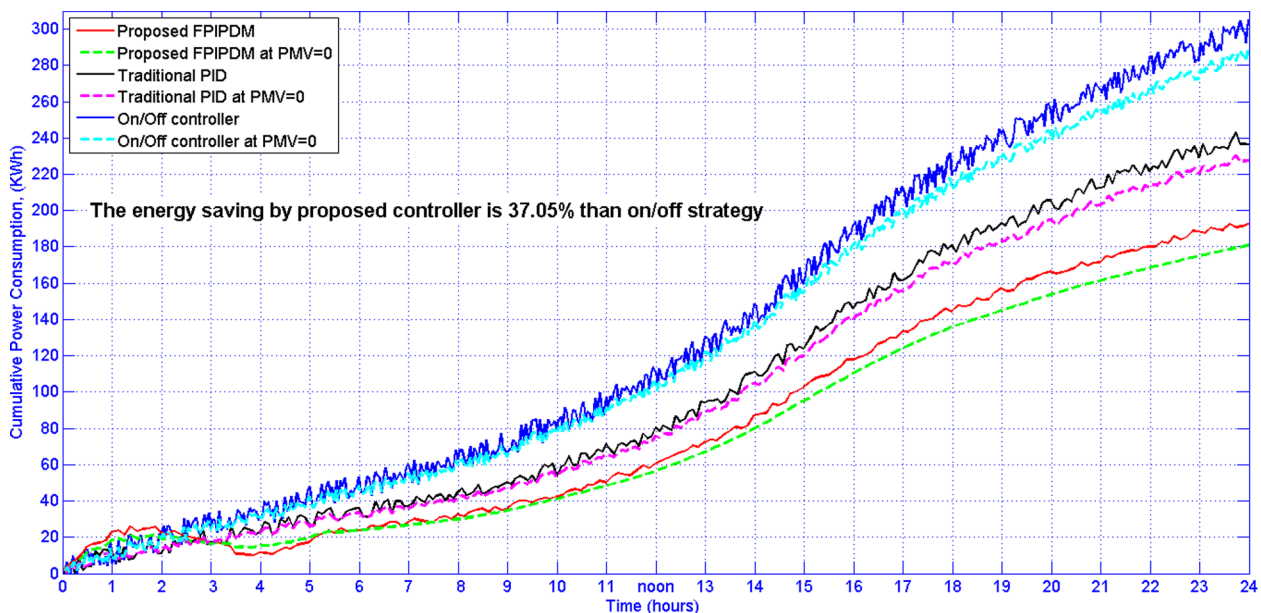


Fig. 23. The results of the cumulative power consumption of the three controllers.

9. Conclusions

A combined two controllers give improved properties when both large and small shifts are to be detected. The fuzzy PI-PD Mamdani-type (FPIPDM) sensitive to parameters uncertainty (parameter variations), which is in contrast to the cluster adaptive based on Takagi-Sugeno-Kang (CABTSK) controller do not need precise parameters of the system and able to minimize the steady-state error due to the different schemes for the two controller's system. The online tuning to spans of the fuzzy membership functions of the FPIPDM cause faster and aggressive response. On the other hand, the converting a Takagi-Sugeno-Kang (TSK) inference into memory hybrid layers of CABTSK to pass the tasks of logical operation, numerical burden and time consuming, utilize this feature by using the gradient descent (GD) algorithm to speed up online tuning and to have better accuracy near the set point. So, the main objective of integrated two different types of controllers is attained by being impressive for its quick response and accurate tracking. The

combined strategy is relatively straightforward even when the controller is involved with multi outputs.

The elevated performance achieved by the proposed controller is realized when the HVAC system delegated to provide recommended indoor thermal comfort, thus executed by manipulating the inputs of AHU. The motivation behind writing this paper is highlight the long-time response for the HVAC systems, especially at starting time, and eliminating steady-state errors, which certainly gives a strong impetus to radically improve control performance, which is an important issue in HVAC systems. Furthermore, the proposed controller achieved a reduction of CO₂ emissions by saving more than 37% of energy when compared to the On/Off type.

CRedit authorship contribution statement

Raad Z. Homod: Conceptualization, Methodology. **Khalaf S. Gaeid:** Validation, Investigation. **Suroor M. Dawood:** Software.

Alireza Hatami: Formal analysis. **Khairul S. Sahari:** Writing - original draft.

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